

Equity Return Predictability with the ICAPM *

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ABSTRACT

This paper highlights a positive and significant beta-return relationship in high expected market return states, as suggested by the ICAPM. The ICAPM has strong out-of-sample predictive power for equity returns. As a result, timing strategies exploiting this predictive power have Sharpe ratios about double those of the buy-and-hold strategies, alphas of about 5% per annum, and average returns increasing sharply with unconditional betas. Our findings relate to the positive beta-return relation uncovered overnight, on macroeconomic announcement days, and in low inflation times because these periods share an important common feature: high market returns.

JEL Classification: D53, G11, G12

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1. Introduction

Predicting returns has always been of particular interest to finance researchers and practitioners because return forecasts dictate trading and risk management strategies. The empirical finance literature has provided ample evidence of market return predictability.¹ A natural way to relate market return predictability to individual asset return predictability is via the intertemporal capital asset pricing model (ICAPM) of [Merton \(1973\)](#), which extends the one-period capital asset pricing model of [Sharpe \(1964\)](#), [Lintner \(1965a,b\)](#), and [Mossin \(1966\)](#) to multiple periods. Under fairly general assumptions, the ICAPM shows that the expected next-period return of an asset is equal to the sum of an intertemporal hedging component and the asset beta times the expected next-period return of the market.²

The ICAPM suggests that the intertemporal hedging component is the dominant determinant of the expected asset return in low expected market return states. However, beta is expected to be the dominant driver of asset returns in high expected market return states. Therefore, one should observe a positive beta-return relation in the data when the market return is expected to be high, but not necessarily when the market return is expected to be low. In the data, we find a positive and significant beta-return relation in high expected market return states, and a significantly weaker relation in low expected market return states. In addition, the sum of an intertemporal hedging component and beta times the expected market

¹Evidence of market return predictability is provided in [Cochrane \(2008\)](#), [Campbell and Thompson \(2008\)](#), [Bollerslev, Tauchen, and Zhou \(2009\)](#), [Rapach, Strauss, and Zhou \(2010\)](#), [Huang, Jiang, Tu, and Zhou \(2015\)](#), [Rapach, Ringgenberg, and Zhou \(2016\)](#), [Martin \(2016\)](#), [Golez and Koudijs \(2018\)](#), [Jiang, Lee, Martin, and Zhou \(2019\)](#), and [Chabi-Yo and Loudis \(2020\)](#) among others. Refer to [Rapach and Zhou \(2013\)](#) for a survey.

²[Graham and Harvey \(2001\)](#) show the CAPM is widely used in the financial industry to compute the cost of capital. [Berk and van Binsbergen \(2016\)](#) and [Barber, Huang, and Odean \(2016\)](#) provide evidence that investments in mutual funds rely on CAPM-alpha. [Ben-David, Li, Rossi, and Song \(2022\)](#), however, challenge these results by showing that simple performance-chasing best explains mutual fund flows. [Harvey and Liu \(2021\)](#) and [Hasler and Martineau \(2023\)](#) show that the market factor is a particularly important driver of asset returns in the time series.

return significantly predicts asset returns out-of-sample. As a result, investment strategies exploiting such predictive power have large Sharpe ratios and risk-adjusted returns (alphas) of about 5% per year.

To predict future asset returns out-of-sample using the ICAPM, the expected next-period market return needs to be estimated using past data only. With the recent surge of papers finding predictors of market returns, we follow [Rapach et al. \(2010\)](#) and use a combination of predictors to forecast the next-period market return. [Rapach et al. \(2010\)](#) show that combining individual forecasts improves information content, stabilizes individual predictive regression model forecasts of the market return, and lowers the forecast variance relative to any of the individual predictive regression model forecasts. The variance risk premium ([Bollerslev et al., 2009](#)), short interest ([Rapach et al., 2016](#)), investor sentiment ([Huang et al., 2015](#)), manager sentiment ([Jiang et al., 2019](#)), and investor attention ([Chen, Tang, Yao, and Zhou, 2022](#)) represent the set of market return predictors, as they have been shown to forecast future market returns with high out-of-sample R^2 and allow to test predictability out-of-sample over a long period of about 58 years. Past market returns are first regressed on each lagged predictor. The fitted values of these five separate regressions provide five estimates of the next-period market return. Aggregating these five estimates using the median, mean, and weights implied by the first principal component yields empirical forecasts of the next-period market return. The main test assets are beta-sorted portfolios, but individual stocks and international portfolios are considered as well. Conditional betas are estimated using conventional rolling regressions of past asset returns on past market returns. To summarize, both the next-period expected market return and asset betas are estimated using past data only, making them suitable to perform out-of-sample analyses.

Using pooled regressions, we first show that betas positively and significantly predict

next-period asset returns when the expected market return is above its trailing median. That is, there is a positive beta-return relation when the expected market return is high. This empirical observation is consistent with the ICAPM prediction that the loading on beta is the expected market return, thereby making the impact of beta on the asset return particularly pronounced in high expected market return states. Furthermore, the beta-return relation observed in low expected market return states is significantly less pronounced than that observed in high expected market return states.

Following the ICAPM, the out-of-sample asset return forecast is the sum of an intertemporal hedging component and the asset beta times the expected market return. As it is the case for the expected market return and the asset betas, the intertemporal hedging component is estimated using past data only. The out-of-sample predictive power of the ICAPM is obtained by computing the out-of-sample R^2 (OOS- R^2) for each test asset, as in [Welch and Goyal \(2007\)](#) and [Campbell and Thompson \(2008\)](#) among others. The OOS- R^2 measures the model's ability to outperform the benchmark historical mean at predicting returns out-of-sample. A positive (resp., negative) OOS- R^2 means the predictive model performs better (resp., worse) than the historical mean. The monthly OOS- R^2 are about 1% at the 1-month forecasting horizon, 5% at the 3-month and 6-month forecasting horizons, and 3% at the 12-month forecasting horizon. The OOS- R^2 s are economically large, statistically significant, and of comparable magnitudes across the different beta-sorted portfolios.

To exploit the positive beta-return relation in periods of high expected market returns, we build a simple trading strategy that invests 100% in the asset when the expected next-period market return is high, and 100% in the risk-free asset otherwise. As expected, the average return of this strategy increases with the beta of the asset in which it invests. That is, despite the fact that low-beta assets deliver higher returns than high-beta assets

unconditionally ([Frazzini and Pedersen, 2014](#)), high-beta assets deliver far larger returns than low-beta assets when conditioning on high expected market returns, which is what this simple strategy exploits. In addition, the Sharpe ratio of the simple strategy is about double that obtained by buying and holding the asset, and its risk-adjusted return (alpha) is about 5% per annum.

We then consider the investment problem faced by a mean-variance investor who has a single-period horizon and invests in both a risky asset and a risk-free asset. Since the ICAPM predicts the risky asset return, the Sharpe ratio of the mean-variance strategy is larger than that of the buy-and-hold strategy. The Sharpe ratios of the mean-variance strategies investing in high-beta portfolios are particularly high relative to those obtained by buying and holding, with multiples of about two. Mean-variance strategies also have large and statistically significant risk-adjusted returns (alphas), ranging from about 2.4% to 4.8% per annum, depending on the beta-sorted portfolio in which they invest and the aggregation method used to compute the expected market return.

Finally, we relate our results to the existing positive beta-return relations uncovered over specific periods such as overnight ([Hendershott, Livdan, and Rösch, 2020](#)), on macroeconomic announcement days ([Savor and Wilson, 2014](#)), on FOMC announcement days ([Ben-Rephael, Carlin, Da, and Israelsen, 2021](#)), and in low inflation months ([Cohen, Polk, and Vuolteenaho, 2005](#)). We show that all of these specific periods feature particularly high market returns. This together with the positive beta-return relation found in high expected market return states suggests that the positive beta-return relation uncovered overnight, on macroeconomic announcement days, on FOMC announcement days, and in low inflation periods might be driven by an important feature common to all of these subsample periods: high market returns.

This paper contributes to the return predictability and investment literature. [Dow \(1920\)](#), [Fama and French \(1988\)](#), and [Campbell and Shiller \(1988\)](#) were among the first to examine stock market return predictability. [Welch and Goyal \(2007\)](#) show that most market return predictors documented in the literature have no or only negligible out-of-sample predictive power. By imposing theoretically motivated restrictions on the regressions performed in [Welch and Goyal \(2007\)](#), [Campbell and Thompson \(2008\)](#) show that the out-of-sample predictive power of the market return predictors improves substantially. [Rapach et al. \(2010\)](#) show that combining the market return forecasts of [Welch and Goyal \(2007\)](#) improves the out-of-sample predictive power. [Cochrane \(2008\)](#) and [van Binsbergen and Koijen \(2010\)](#) argue and show that market returns are positively and significantly predicted by dividend yields. [Golez and Koudijs \(2018\)](#) collect annual stock market return data over four centuries and find that dividend yields consistently forecast market returns. [Rapach, Strauss, and Zhou \(2013\)](#) provide evidence that U.S. market returns predict future market returns of various international markets. [Bali and Engle \(2010\)](#) use a dynamic conditional correlation model to estimate conditional covariances between stock returns and the market. As predicted by the ICAPM, the conditional covariances, which reflect investors' myopic demands for stocks, positively and significantly forecast future stock returns. When predicting market returns out-of-sample, [Dangl and Halling \(2012\)](#) show that predictive regressions with time-varying coefficients significantly outperform predictive regressions with constant coefficients. [Rossi and Timmermann \(2015\)](#) estimate the conditional covariance between the market return and a measure of time-varying investment opportunities. As predicted by the ICAPM, the estimated conditional covariance, which reflects investors' hedging motives, forecasts future market returns both in- and out-of-sample. [Gargano, Pettenuzzo, and Timmermann \(2019\)](#) show that a three-factor model significantly predicts bond returns

out-of-sample, and that trading strategies exploiting such predictive power have high risk-adjusted returns. [Sabbatucci \(2022\)](#) shows that, when M&A cash dividends are taken into account, the dividend-price ratio predicts future market returns significantly better than when they are ignored. [Barroso and Santa-Clara \(2015\)](#) show that scaling the momentum strategy by a measure of its current volatility doubles the strategy's Sharpe ratio. Similarly, [Moreira and Muir \(2017\)](#) show that scaling trading strategies such as value, momentum, profitability, investment, and betting-against-beta by a measure of their current volatility generates large Sharpe ratios and risk-adjusted returns. [Cederburg, O'Doherty, Wang, and Yan \(2020\)](#) show that the volatility-managed strategies presented in [Moreira and Muir \(2017\)](#) are not implementable in real time, and that the out-of-sample versions of them generate lower Sharpe ratios and certainty equivalent returns than the original unmanaged strategies.

In recent years, several other studies have proposed new return predictors. [Bollerslev et al. \(2009\)](#) and [Pyun \(2019\)](#) show that the variance risk premium is a powerful predictor of future market returns. [Della Corte, Ramadorai, and Sarno \(2016\)](#) provide evidence that currency volatility risk premia predict future exchange rate returns. [Huang et al. \(2015\)](#), [Rapach et al. \(2016\)](#), [Jiang et al. \(2019\)](#), and [Chen et al. \(2022\)](#) show that investor sentiment, short interest, manager sentiment, and investor attention, respectively, are powerful predictors of market returns. [Martin \(2016\)](#) and [Martin and Wagner \(2019\)](#) show that the SVIX predicts market returns and individual S&P 500 stock returns, respectively. [Kadan and Tang \(2020\)](#) provide a lower bound for the expected return of a stock and show it strongly, positively, loads on the beta of the stock. [Rossi \(2018\)](#) and [Gu, Kelly, and Xiu \(2020\)](#) provide evidence that machine learning techniques are particularly valuable to predict portfolio returns and individual stock returns out-of-sample.

Our paper contributes to the existing literature by providing evidence that asset returns

and betas are positively related when market returns are expected to be high. This likely explains the positive beta-return relation uncovered overnight (Hendershott et al., 2020), on macroeconomic announcement days (Savor and Wilson, 2014), on FOMC announcement days (Ben-Rephael et al., 2021), and in low inflation times (Cohen et al., 2005) because all of these subsample periods feature particularly high market returns. We also show that the ICAPM has strong out-of-sample predictive power for future asset returns. Consequently, trading strategies exploiting such out-of-sample predictive power feature large Sharpe ratios and substantial risk-adjusted returns (alphas).

The remainder of the paper is organized as follows. Section 2 provides the theoretical guidance used in our empirical analysis by summarizing the ICAPM framework. Section 3 describes the data, highlights the relation between asset returns and betas conditional on the level of the expected market return, and documents the out-of-sample predictive power of the ICAPM. Section 4 discusses the performance of trading strategies exploiting out-of-sample return predictability. Section 5 investigates the robustness of the results. Section 6 relates our findings to the existing positive beta-return relation uncovered overnight, on macroeconomic announcement days, and in low inflation times. Section 7 concludes. Derivations and supplementary material are provided in the Internet Appendix.

2. Theoretical Guidance: ICAPM

This section summarizes the Intertemporal Capital Asset Pricing Model (ICAPM) developed by Merton (1973), and therefore provides a theoretical guidance for the empirical results that follow.

The ICAPM features few assumptions, which makes it a fairly general and insightful equilibrium model. The model features no transactions costs or taxes. There are $n +$

1 infinitely divisible assets (n risky and one risk-free) that are traded continuously by k investors who choose their consumption plan and portfolio to maximize their expected time-additive utility of lifetime consumption and terminal wealth. The vector of n risky asset prices has the following dynamics

$$dS_t = \mathbf{I}_{S_t} \mu(Y_t, t) dt + \mathbf{I}_{S_t} \Sigma(Y_t, t) dB_t,$$

where $\mu(Y_t, t) \equiv \mu_t$ is the n -dimensional vector of conditional expected returns, $\Sigma(Y_t, t) \equiv \Sigma_t$ is the $n \times n$ matrix of conditional return diffusion, $\mathbf{I}_{S_t}$ is the $n \times n$ diagonal matrix of risky asset prices, and Y_t is an m -dimensional vector of state variables driving the expected return vector, the diffusion matrix, and the risk-free rate $r_f(Y_t, t) \equiv r_{ft}$ paid by the risk-free asset at time t .

As shown in [Merton \(1973\)](#), aggregating investors risky asset demands and requiring market clearing (i.e., aggregate demand equals aggregate supply) yields the following characterization of conditional expected excess asset returns

$$\mu_{it} - r_{ft} = \beta_{it}(\mu_{Mt} - r_{ft}) + \underbrace{\sum_{j=1}^m \beta_{it}^{Hj}(\mu_{Hjt} - r_{ft})}_{\equiv c_{it}}, \quad i \in \{1, \dots, n\} \quad (1)$$

where μ_M is the conditional expected return of the market portfolio, μ_{Hj} is the conditional expected return of hedging portfolio j , and β_i and β_i^{Hj} are the conditional exposures of asset i 's return to the market return and hedging portfolio j 's return, respectively. Discretizing

Equation (1) yields

$$\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h}) + \underbrace{\sum_{j=1}^m \beta_{it}^{Hj}\mathbb{E}_t(r_{Hj,t+h})}_{\equiv c_{i,t+h}}, \quad (2)$$

where h is the forecasting horizon, $r_{i,t+h}$ is the h -period-ahead excess return of asset i , $r_{M,t+h}$ is the h -period-ahead excess return of the market, $r_{Hj,t+h}$ is the h -period-ahead excess return of hedging portfolio j , and $\mathbb{E}_t(\cdot)$ denotes the time- t conditional expectation.

The ICAPM in (1) and (2) shows that the expected excess return of any asset depends on two components. The first component is the product of the asset's exposure to the market, *beta* (β), multiplied by the expected excess return of the market. This first component—explicit beta-component—is explicitly and positively driven by beta because the excess market return is expected to be positive. The second component, called the *intertemporal hedging component*, is a sum of the asset's exposures to the hedging portfolios multiplied by the corresponding expected excess returns of the hedging portfolios. This second component is not explicitly driven by beta, but could potentially be implicitly driven by it. Note that the intertemporal hedging component, $c_{i,t+h}$, is a weighted sum of expected h -period-ahead excess returns, which explains its dependence on the forecasting horizon h .

Equations (1) and (2) show that the intertemporal hedging component is the dominant driver of the cross-section of expected asset returns when the expected return of the market is low, or in other words, close to zero. Since the implicit dependence of the intertemporal hedging component on beta is unclear, the beta-return relation one should observe in the data in low expected market return states is ambiguous. When the expected market return is high, the explicit beta-component per se predicts a positive and strong beta-return relation. Consequently, one should observe in the data a positive beta-return relation in high

expected market return states, unless the intertemporal hedging component of high-beta (resp., low-beta) assets is strongly negatively (resp., positively) correlated to the expected market return. It is also worth noting that if the data-generating process is not the ICAPM but the conditional CAPM (i.e., $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$), then one should observe a positive beta-return relation in high expected market return states and no significant relation in low expected market return states. In addition, the ICAPM suggests that the sum of the intertemporal hedging component and beta times the expected market return should predict future asset returns in the data. Of course, these facts should be empirically observed provided that beta, the expected market return, and the intertemporal hedging component are estimated with reasonable accuracy. Section 3 presents the empirical tests and the details on how beta, the expected market return, and the intertemporal hedging component are estimated.

3. Beta-Return Relation and Return Predictability

In this section, we describe the data and the empirical methodology. We show that asset betas predict positively and significantly asset returns in high expected market return states. In addition, the sum of the intertemporal hedging component and the product of beta and the expected market return predicts asset returns out-of-sample, as suggested by the ICAPM.

3.1. Data

We retrieve excess market returns, r_{Mt} , from Kenneth French's website.³ Following [Black, Jensen, and Scholes \(1972\)](#) and [Jylha \(2018\)](#) among others, we construct ten monthly beta-

³Data source: <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french>

sorted portfolios using U.S. common stocks that are identified in CRSP as having a share code of 10 or 11. Monthly stock betas are the slope estimates obtained from a rolling regression of the past 24 monthly excess stock returns on excess market returns. At the beginning of each month, stocks are sorted into one of the ten beta-deciles, and the monthly value-weighted excess return, r_{it} , of each decile portfolio i is computed over the month. For each portfolio, we estimate monthly rolling betas, β_{it} , by regressing the past 24 monthly portfolio excess returns on the past 24 monthly excess market returns. The ten beta-sorted portfolios represent our main test assets. In Sections 4.3 and 5, we also consider individual stocks and international portfolios for robustness.

3.2. Methodology

In order to forecast excess market returns out-of-sample, we follow [Rapach et al. \(2010\)](#) and use a combination approach. In recent years, several predictors were proposed as good predictors of excess market returns. [Rapach et al. \(2010\)](#) show that combining individual forecasts improves information content, stabilizes individual predictive regression model forecasts of the market return, and lowers the forecast variance relative to any of the individual predictive regression model forecasts. It also allows to obtain a fairly long out-of-sample period. We select the following predictors: variance risk premium ([Bollerslev et al., 2009](#)), short interest ([Rapach et al., 2016](#)), investor sentiment ([Huang et al., 2015](#)), manager sentiment ([Jiang et al., 2019](#)), and investor attention ([Chen et al., 2022](#)). We choose these predictors because their original studies show they have high out-of-sample R^2 at predicting excess market return. The variance risk premium data is retrieved from the personal website of Hao Zhou's personal website⁴ and the remaining predictors are retrieved from Guofu

⁴<https://sites.google.com/site/haozhouspersonalhomepage/>

Zhou's personal website⁵.

The sample periods over which the predictors are available vary. The variance risk premium is available from January 1990 to December 2022. The short-interest index is available from January 1973 to December 2021. The investor sentiment index is available from July 1965 until December 2020. The manager sentiment index is available from January 2003 to December 2017. The investor attention index is available from 1980 to December 2017. By combining these individual predictors, our sample period extends from July 1965 to December 2022, for approximately 58 years.

We begin with a standard predictive regression model for the excess market return, expressed as

$$r_{M,\tau} = a_t^i + b_t^i x_{\tau-h}^i + \epsilon_\tau, \quad \tau = 1, \dots, t \quad (3)$$

where r_M is the excess market return, x^i is predictor i , and h is the forecasting horizon. Out-of-sample forecasts of the excess market return are generated using an expanding estimation window, characterized by the expanding time index t in Equation (3). For each predictor, we select a 10-year burn-in (in-sample) period. The out-of-sample forecast of the h -month-ahead excess market return based on predictor x^i is given by

$$\hat{r}_{M,t+h}^i = \hat{a}_t^i + \hat{b}_t^i x_t^i, \quad (4)$$

where $\hat{a}_t^i$ and $\hat{b}_t^i$ are the OLS estimates of a_t^i and b_t^i , respectively, over the expanding window described in Equation (3). Following this procedure through the end of the out-of-sample period, we generate a time series of out-of-sample forecasts of the excess market return for each predictor x^i . Inspired by Rapach et al. (2010), the final out-of-sample h -month-

⁵<http://apps.olin.wustl.edu/faculty/zhou/>

ahead excess market return forecast, $\mathbb{E}_t(r_{M,t+h})$, is obtained by aggregating the out-of-sample forecasts, $\hat{r}_{M,t+h}^i$, defined in Equation (4) using three different methods:

1. Median aggregation method:

$$\mathbb{E}_t(r_{M,t+h}) = \text{median}(\hat{r}_{M,t+h}^1, \hat{r}_{M,t+h}^2, \dots, \hat{r}_{M,t+h}^N) \quad (5)$$

2. Mean aggregation method:

$$\mathbb{E}_t(r_{M,t+h}) = \frac{1}{N} \sum_{i=1}^N \hat{r}_{M,t+h}^i \quad (6)$$

3. PCA aggregation method:

$$\mathbb{E}_t(r_{M,t+h}) = \sum_{i=1}^N \omega_{it} \hat{r}_{M,t+h}^i \quad (7)$$

N is the number of predictors, $\text{median}(\cdot)$ is the median operator, PCA stands for Principal Component Analysis, and ω_{it} is the time-varying weight of forecast $\hat{r}_{M,t+h}^i$ implied by the first principal component, computed over a 2-year historical moving window, of all currently available forecasts. If there is only one predictor available, we impose a weight of 100%. For each predictor, we require a 10-year initial burn-in period to compute out-of-sample forecasts. We then require an additional 2-years to compute the PCA weights from the out-of-sample forecasts. Because the earliest available predictor is available from July 1965, with the initial burn-in period is 10 years and the required 2-year window for computing the PCA weights, the out-of-sample period is from July 1977 to December 2022. The simple median and mean of the individual forecasts are also computed from July 1977 to keep the sample constant across the various aggregation methods.

Table 1 reports the out-of-sample R^2 (OOS- R^2), as measured in [Welch and Goyal \(2007\)](#) and [Campbell and Thompson \(2008\)](#), for different forecasting horizons $h \in \{1, 3, 6, 12\}$. Columns (1)–(2), (3)–(4), and (5)–(6) show the OOS- R^2 for the median, mean, and PCA aggregation methods for the expected market return, respectively. Columns (2), (4), and (6) impose the restriction that the expected market return is non-negative ($\mathbb{E}_t(r_{M,t+h}) \geq 0$), as in [Campbell and Thompson \(2008\)](#) and [Pettenuzzo, Timmermann, and Valkanov \(2014\)](#). A positive OOS- R^2 indicates that the market return forecast, $\mathbb{E}_t(r_{M,t+h})$, predicts future market returns better than the historical mean, whereas a negative value means that the forecast is a worse predictor than the historical mean. As emphasized in [Welch and Goyal \(2007\)](#), beating the historical mean is particularly difficult when predicting future market returns.

When no restrictions on the expected market return are imposed, the OOS- R^2 are positive and statistically significant at all horizons for the median aggregation method. For the mean aggregation method, the OOS- R^2 are positive and statistically significant at the 3-month, 6-month, and 12-month horizons when no restrictions are imposed. Not restricting the expected market return when using the PCA aggregation method yields positive and statistically significant OOS- R^2 at the 6-month and 12-month forecasting horizons. Imposing the restriction that the expected market return is non-negative implies that the OOS- R^2 are positive and statistically significant at the 1% level for all forecasting horizons and aggregation methods. Imposing this restriction increases substantially the OOS- R^2 obtained with the mean and PCA aggregation methods. Indeed, at the 3-month, 6-month, and 12-month forecasting horizons, both the mean and PCA aggregation methods yield particularly high OOS- R^2 , ranging between 5% and 10.7% when the restriction is imposed. Imposing the restriction, however, has a minor impact on the OOS- R^2 obtained using the median

Table 1
Forecasting Excess Market Returns Out-of-Sample

This table reports the out-of-sample R^2 (OOS- R^2) computed as follows

$$\text{OOS-}R^2 = 1 - \frac{\sum_{t=1}^N (r_{M,t+h} - \mathbb{E}_t(r_{M,t+h}))^2}{\sum_{t=1}^N (r_{M,t+h} - \bar{r}_{M,t+h})^2},$$

where $r_{M,t+h}$ is the excess market returns over horizon h , $\bar{r}_{M,t+h}$ is the average historical excess market return over horizon h prior to month t , and $\mathbb{E}_t(r_{M,t+h})$ is the expected excess market return defined in Equations (5), (6), and (7). Columns (1)-(2), (3)-(4), and (5)-(6) show the OOS- R^2 for the median, mean, and PCA aggregated forecasts, respectively. Columns (2), (4), and (6) impose the restriction that $\mathbb{E}_t(r_{M,t+h}) \geq 0$. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Restriction: $\mathbb{E}_t(r_{M,t+h}) \geq 0$	Aggregation method for the expected market return					
	Median		Mean		PCA	
	No (1)	Yes (2)	No (3)	Yes (4)	No (5)	Yes (6)
$h = 1$	0.014***	0.012***	0.007	0.014***	-0.028	0.017***
$h = 3$	0.041***	0.036***	0.040***	0.050***	-0.025***	0.076***
$h = 6$	0.049***	0.040***	0.065***	0.070***	0.023***	0.107***
$h = 12$	0.031***	0.024***	0.061***	0.062***	0.052***	0.102***

aggregation method.⁶ Overall, Table 1 shows that the market return forecast predicts future market returns reasonably well, and that imposing a non-negativity restriction on the market return forecast tends to improve out-of-sample predictive power ([Campbell and Thompson, 2008](#)). For this reason, the non-negativity restriction on the expected market return is imposed from now on.

⁶The predictive power of the individual market return forecasts are compared to the predictive power of the aggregated market return forecasts in the Internet Appendix (see Section IA.2) by plotting cumulative squared prediction errors as in [Welch and Goyal \(2007\)](#).

3.3. Asset Returns versus Beta in Different Expected Market Regimes

To show that asset betas predict more strongly and positively asset returns when market returns are predicted to be high than when they are predicted to be low, we estimate the following pooled regression⁷

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h}, \quad (8)$$

where h is the forecasting horizon, $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2, and β_{it} is the slope coefficient obtained by regressing the past 24 monthly excess returns of portfolio i on the past 24 monthly excess returns of the market. The dummy variable $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ equals one if the expected market return, $\mathbb{E}_t(r_{M,t+h})$, is below its trailing median and zero otherwise. That is, the cutoff value determining whether the current expected market return is considered high or low is the median of the historical empirical distribution of the expected market return. We believe that this is the most intuitive, parsimonious, and natural way of determining whether a value is considered high or low. In Internet Appendix IA.4.3, we show that the results presented in this section are robust to alternative definitions of the high-low cutoff value for the expected market return.

Table 2 reports the output of regression (8) for different forecasting horizons and aggregation methods used to compute the expected market return. Standard errors are clustered by time to adjust for the cross-sectional correlation of the residuals. The loading on β_{it} , b_1 , measures the slope of the beta-return relation when the expected market return is above its trailing median. b_1 is positive and statistically significant for all forecasting horizons and aggregation methods used to compute the expected market return. That is, there

⁷Pooled and panel regressions in empirical asset pricing have been advocated by [Petersen \(2008\)](#) and used in [Savor and Wilson \(2014\)](#), [Martin and Wagner \(2019\)](#), and [Kremens and Martin \(2019\)](#) among others.

is a positive beta-return relation in high expected market return states. The loading on $\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$, b_2 , measures the change in the slope of the beta-return relation when entering low expected market return states. b_2 is negative and statistically significant, meaning that the beta-return relation is significantly weaker in low expected market return states than in high expected market return states. Furthermore, the sum of the loadings b_1 and b_2 measures the slope of the beta-return relation in low expected market return states. This sum is negative and statistically significant (p -values are provided in the last row of Table 2) for all forecasting horizons and aggregation methods used to compute the expected market return. That is, the beta-return relation is significantly negative in low expected market return states. This result shows that the conditional CAPM (i.e., $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$), which is a restricted version of the ICAPM described in Equation (2), is unlikely to be the true data-generating process. Indeed if that were the case, the beta-return relation observed in low expected market return states should be insignificant. Overall, the results presented in Table (8) show the importance of conditioning on expected market returns, as the ICAPM suggests (see Section 2), in order to highlight the existing relation between asset returns and beta.⁸ Without conditioning on expected market return, there is no relation between asset returns and beta (see, for instance, [Black et al., 1972](#), [Fama and French, 1992](#), [Savor and Wilson, 2014](#), [Jylha, 2018](#)). In other words, the security market line is flat unconditionally, but it is significantly upward-sloping when future market returns are expected to be high.

⁸The impact of considering a sentiment-based forecast of the market return (i.e., aggregation of the individual forecasts obtained with investor sentiment, managerial sentiment, and investor attention) versus a price-based forecast of the market return (i.e., aggregation of the individual forecasts obtained with the variance risk premium and short-interest) on the conditional beta-return relation is discussed in Internet Appendix IA.6.

Table 2
Pooled Regression: High and Low Expected Market Return

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return, $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns, and $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below its trailing median and zero otherwise. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected excess market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. The last row of the table reports the p -value for the null: $b_1 + b_2 = 0$. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.006*** (0.001)	0.019*** (0.003)	0.032*** (0.005)	0.044*** (0.008)	0.005*** (0.001)	0.017*** (0.003)	0.025*** (0.006)	0.036*** (0.007)	0.009*** (0.002)	0.018*** (0.004)	0.047*** (0.005)	0.069*** (0.008)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.013*** (0.001)	-0.042*** (0.009)	-0.070*** (0.013)	-0.098*** (0.022)	-0.013*** (0.001)	-0.045*** (0.002)	-0.074*** (0.006)	-0.136*** (0.033)	-0.015*** (0.002)	-0.029*** (0.003)	-0.084*** (0.010)	-0.151*** (0.035)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	0.000 (0.001)	-0.002 (0.009)	-0.007 (0.013)	-0.015 (0.024)	-0.000 (0.001)	-0.002 (0.002)	-0.006 (0.005)	-0.017 (0.033)	0.005*** (0.002)	-0.004 (0.003)	0.027*** (0.009)	0.038 (0.033)
Intercept	0.005*** (0.001)	0.014*** (0.004)	0.033*** (0.007)	0.081*** (0.014)	0.006*** (0.002)	0.017*** (0.004)	0.039*** (0.007)	0.096*** (0.012)	0.003 (0.002)	0.017*** (0.004)	0.021*** (0.007)	0.069*** (0.012)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.014	0.050	0.069	0.069	0.015	0.055	0.072	0.114	0.012	0.033	0.057	0.096
$p\text{-val. } b_1 + b_2 = 0$	0.000	0.002	0.002	0.015	0.000	0.000	0.000	0.007	0.000	0.000	0.000	0.009

3.4. Out-of-Sample Predictability

Providing evidence that the ICAPM can be used to predict asset returns out-of-sample is particularly important because it implies investors can use the ICAPM to build real-time investment strategies.

The ICAPM in Equation (2) states that the sum of an intertemporal hedging component and the product of the asset beta and the expected market return predicts future asset returns:

$$\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h}) + c_{i,t+h}. \quad (9)$$

To construct out-of-sample portfolio return forecasts that rely on Equation (9), we compute out-of-sample estimates of expected market returns and betas as in the previous sections (see Sections 3.1 and 3.2). By definition, the intertemporal hedging component is the difference between the expected asset return and beta times the expected market return:

$$c_{i,t+h} = \mathbb{E}_t(r_{i,t+h}) - \beta_{it}\mathbb{E}_t(r_{M,t+h}).$$

Consequently, a natural empirical estimate of the intertemporal hedging component, $c_{i,t+h}$, is the historical average of the difference between portfolio i return $r_{i\tau}$ and the product $\beta_{i,\tau-h}\mathbb{E}_{\tau-h}(r_{M,\tau})$ with $\tau = 1, \dots, t$. That is, $c_{i,t+h}$ is estimated over an expanding window and is therefore updated every month.⁹ As in the previous section, the out-of-sample period is from July 1977 to December 2022.

Table 3 reports the OOS- R^2 obtained by predicting portfolio i return with the forecast

⁹Rolling estimations of the intertemporal hedging component and their impact on portfolio return predictability are discussed in Section 5.2.

described in Equation (9) for different forecasting horizons and aggregation methods for the expected market return. At the 1-month horizon, the OOS- R^2 ranges from 0.6% to 1.6%. The OOS- R^2 are particularly large at the 3-month and 6-month horizons, with values ranging from 1.3% to 14.1%. For these forecasting horizons, the mean and PCA aggregation methods provide the largest OOS- R^2 . The OOS- R^2 are not only large but also highly statistically significant. At the 3-month and 6-month horizons, the OOS- R^2 are statistically significant at the one percent level for all beta-sorted portfolios and aggregation methods. Although all beta-sorted portfolios feature statistically significant out-of-sample return predictability at the 3-month and 6-month horizons, the return predictability of the bottom five decile portfolios tends to be slightly less pronounced than that of the top five decile portfolios.

It is worth noting that the OOS- R^2 obtained for beta-sorted portfolio returns in Table 3 are of the same order of magnitude as those obtained for market returns in Table 1. They are about 1% at the monthly horizon, 5% at the 3-month and 6-month horizons, and 3% at the 12-month horizons. This suggests that the estimated betas and intertemporal hedging components are fairly good proxies for the true betas and intertemporal hedging components, respectively. Indeed, if the estimated betas and intertemporal hedging components were not good proxies, then the gap between the OOS- R^2 obtained for market returns and those obtained for beta-sorted portfolio returns would be substantial. Overall, the results reported in Table 3 provides evidence that the ICAPM can be used to forecast future asset returns out-of-sample.

Table 3
Out-of-Sample R^2 for Beta-Sorted Portfolios

This table reports the out-of-sample R^2 (OOS- R^2) for the ten beta-sorted portfolios. The OOS- R^2 for portfolio i is computed as follows

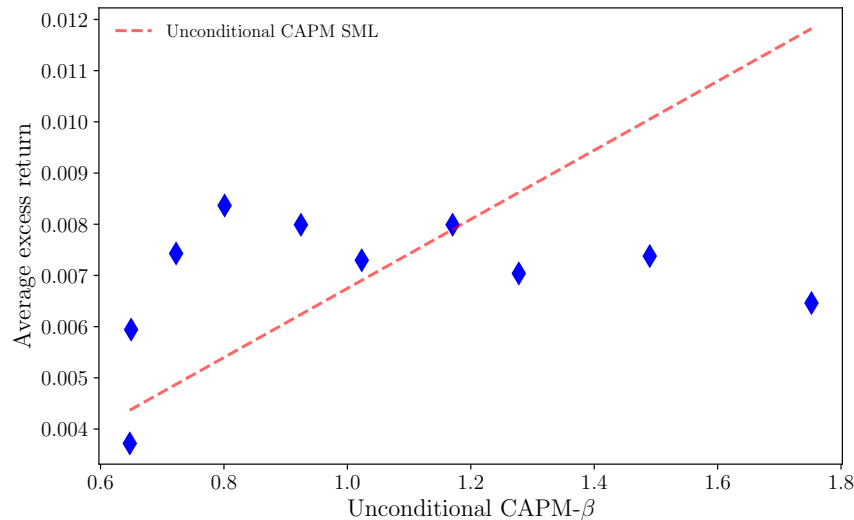
$$\text{OOS-}R_i^2 = 1 - \frac{\sum_{t=1}^N (r_{i,t+h} - \mathbb{E}_t(r_{i,t+h}))^2}{\sum_{t=1}^N (r_{i,t+h} - \bar{r}_{i,t+h})^2},$$

where $r_{i,t+h}$ is portfolio i excess return over horizon h , $\bar{r}_{i,t+h}$ is the average historical return of portfolio i over horizon h prior to month t , and $\mathbb{E}_t(r_{i,t+h})$ for $h = 1, 3, 6, 12$, is the expected return of portfolio i computed as in Equation (9). In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected excess market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.009***	0.030***	0.037***	0.036***	0.008**	0.037***	0.053***	0.059***	0.007**	0.048***	0.062***	0.047***
2	0.006*	0.019***	0.013***	-0.011***	0.006**	0.027***	0.032***	0.013***	0.011***	0.052***	0.059***	0.024***
3	0.008***	0.028***	0.031***	0.023***	0.008**	0.036***	0.050***	0.045***	0.012***	0.062***	0.082***	0.066***
4	0.010***	0.033***	0.040***	0.034***	0.010***	0.042***	0.061***	0.059***	0.015***	0.071***	0.102***	0.096***
5	0.009***	0.033***	0.035***	0.012***	0.010***	0.043***	0.057***	0.035***	0.014***	0.073***	0.102***	0.080***
6	0.010***	0.033***	0.028***	-0.011***	0.011***	0.045***	0.055***	0.020***	0.014***	0.075***	0.101***	0.068***
7	0.010***	0.040***	0.044***	0.014***	0.013***	0.056***	0.077***	0.054***	0.016***	0.085***	0.123***	0.101***
8	0.010***	0.037***	0.030***	-0.033***	0.012***	0.055***	0.066***	0.009***	0.015***	0.083***	0.116***	0.071***
9	0.011***	0.043***	0.043***	-0.009***	0.014***	0.063***	0.083***	0.039***	0.015***	0.088***	0.141***	0.111***
High β	0.008***	0.040***	0.033***	-0.048***	0.012***	0.065***	0.082***	0.011***	0.008***	0.073***	0.129***	0.073***

Figure 1. Security Market Line for the Ten Beta-Sorted Portfolios

This figure shows the unconditional CAPM security market line (red-dashed line) and the scatter plot (blue diamonds) of the ten beta-sorted portfolios average excess returns on their unconditional CAPM betas. The sample period is from July 1977 to December 2022.



4. Trading Strategies

In this section we implement and discuss the performance of trading strategies that exploit the return predictability documented in Sections 3.3 and 3.4. Strategies exploiting the out-of-sample predictive power of the ICAPM feature Sharpe ratios that are up to twice as large as those of the corresponding buy-and-hold strategies, and annualized risk-adjusted returns (alphas) of about 5%.

4.1. Simple Strategy

We start by plotting on Figure 1 the average excess returns obtained by buying and holding the different beta-sorted portfolios against the unconditional CAPM betas of these buy-and-hold strategies (blue diamonds). Consistent with the findings of [Black et al. \(1972\)](#) among

others, the average excess returns of the buy-and-hold strategies investing in low-beta portfolios are similar to those of the buy-and-hold strategies investing in high-beta portfolios. In addition, the differences between the blue diamonds and the red-dashed line, which depicts the CAPM SML, show that low-beta portfolios have much larger risk-adjusted returns (alphas) than high-beta portfolios. Consequently, an investment strategy that bets against beta generates large risk-adjusted returns, as documented by [Frazzini and Pedersen \(2014\)](#). This means that taking high risk does not yield a high average return when the investment strategy (buy-and-hold in this case) does not condition on the level of the expected market return. In contrast and as shown in what follows, implementing a simple timing strategy that conditions on the level of the expected market return does generate a positive risk-return tradeoff.

To understand how a positive risk-return tradeoff can be obtained via market timing, one has to understand the characteristics of the intertemporal hedging component defined in Equation (9). Figure 2 plots the estimated intertemporal hedging component, $c_{i,t+h}$ with $h = 3$, for each beta-sorted portfolio i . The intertemporal hedging component tends to be large for low-beta portfolios and low for high-beta portfolios. This empirical observation together with the ICAPM in Equation (9) imply that low-beta portfolios have higher expected returns than high-beta portfolios when the expected market return is low. Indeed, when the expected market return is close to zero, the intertemporal hedging component is the dominant driver of the portfolio expected return. Because this hedging component is larger for low-beta portfolios than for high-beta portfolios, one observes a negative beta-return relation in the data when the expected market return is low. In contrast, when the expected market return is high, the beta-return relation is positive (see Table 2) because the product of beta and the expected market return becomes the dominant driver of the portfolio expected return.

Consequently, one simply has to invest in the risky asset when the future market return is expected to be high, and in the risk-free asset otherwise, to obtain a positive risk-return tradeoff.

Specifically, the simple timing strategy consists in going long 100% of wealth in portfolio i if the non-negative expected market return, $\mathbb{E}_t(r_{M,t+h}) \geq 0$, defined in Section 3.2 is above its historical expanding median, and long 100% of wealth in the risk-free asset otherwise. Figure 3 depicts the security market lines for the simple trading strategies when considering different forecasting horizons h and different aggregation methods for the expected market return. The aggregation method for the expected market return is the median in Panel A, the mean in Panel B, and the PCA in Panel C. As expected, there is a clear positive relation between the strategy's average excess return and the beta of the portfolio in which it invests. This positive relation is observed for all forecasting horizons and aggregation methods used to compute the expected market return. To summarize, despite the fact that low-beta assets and high-beta assets have similar returns unconditionally ([Black et al., 1972](#)) and that low-beta assets have larger risk-adjusted returns than high-beta assets unconditionally ([Frazzini and Pedersen, 2014](#)), high-beta assets deliver far larger returns than low-beta assets when conditioning on high expected market returns. The latter positive conditional beta-return relation is what the simple strategy presented in this section exploits.

Figure 2. Intertemporal Hedging Component, $c_{i,t+h}$, with $h = 3$ Months

This figure plots the estimated intertemporal hedging component, $c_{i,t+h}$, for each beta-sorted portfolio i , where the forecasting horizon is $h = 3$ months. $c_{i,t+h}$ is defined as the historical average of the difference between portfolio i return $r_{i\tau}$ and the product $\beta_{i,\tau-h}\mathbb{E}_{\tau-h}(r_{M,\tau})$ with $\tau = 1, \dots, t$. $\mathbb{E}_{\tau-h}(r_{M,\tau}) \geq 0$ is the non-negative expected excess market return defined in Section 3.2 and constructed using the PCA aggregation method. The sample period is from July 1977 to December 2022.

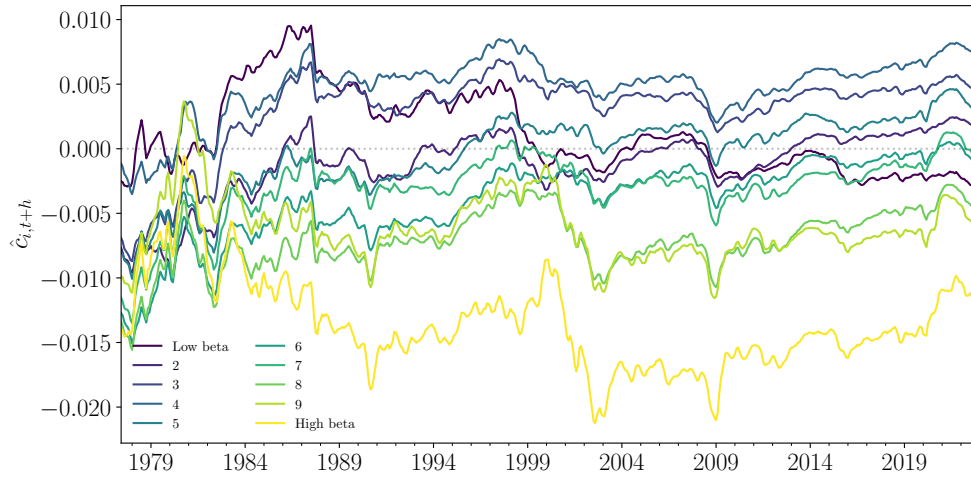
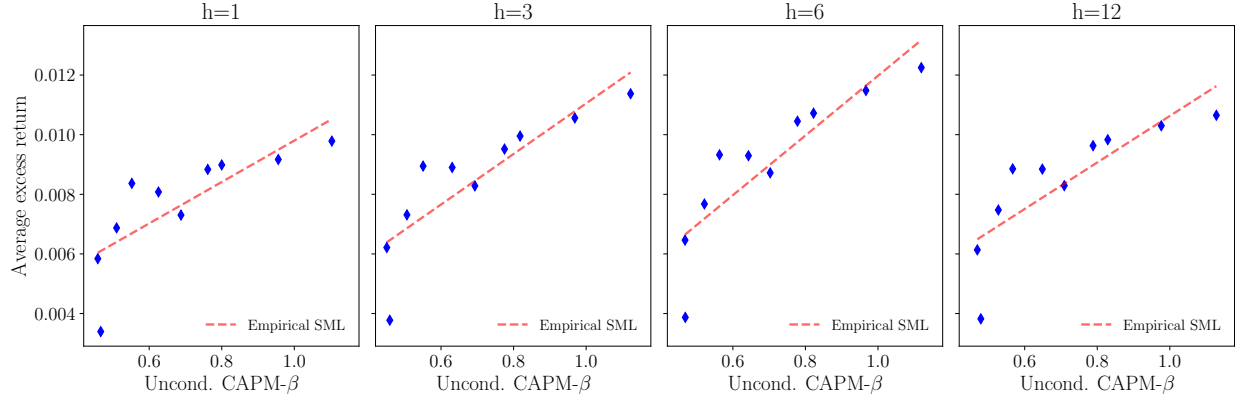


Table 4 reports the monthly CAPM-alphas, monthly Fama-French 5-factor-alphas (FF5-alphas), and annualized Sharpe ratios of the buy-and-hold and simple strategies investing in the ten beta-sorted portfolios, using the 3-month forecasting horizon. The results obtained using the 1-month, 6-month, and 12-month forecasting horizons are similar and are reported in Table IA1 of the Internet Appendix. As usual, the alpha is the intercept obtained by regressing the strategy excess returns on excess market returns for the CAPM-alpha, and on the five Fama-French factors for the FF5-alpha. The Sharpe ratio is the average annualized excess return of the strategy divided by its annualized excess return volatility.

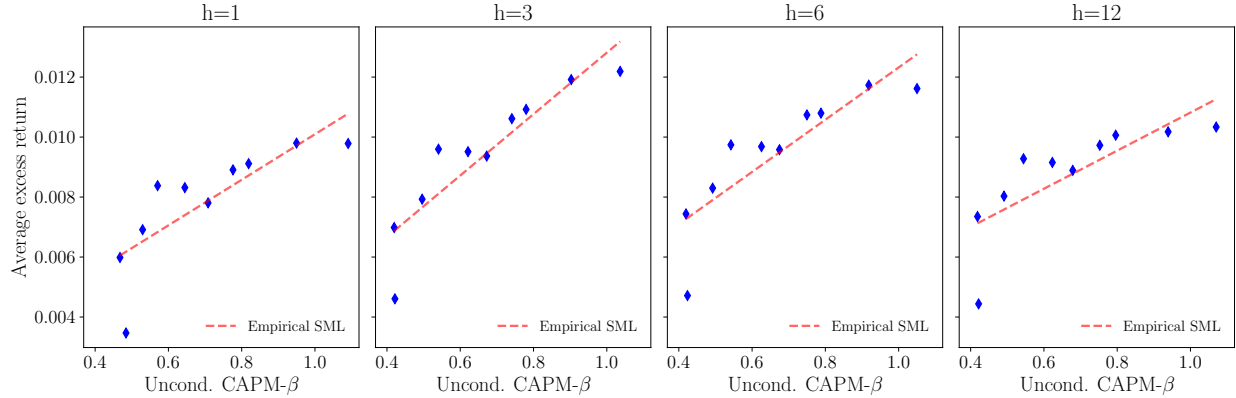
Figure 3. Security Market Line for the Simple Strategies Exploiting the Return Predictability of the Ten Beta-Sorted Portfolios

This figure presents the security market line for trading strategies that invest 100% of wealth in beta-sorted portfolio i when the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is above its historical expanding median, and 100% of wealth in the risk-free asset otherwise. $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods in Panels A to C, respectively. The sample period is from July 1977 to December 2022.

Panel A: Median aggregation method for the expected market return



Panel B: Mean aggregation method for the expected market return



Panel C: PCA aggregation method for the expected market return

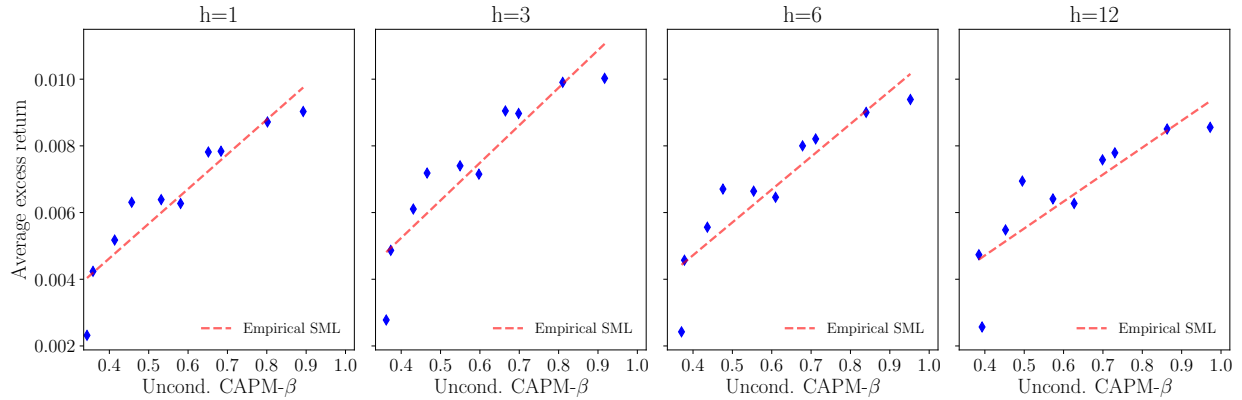


Table 4
Alpha and Sharpe Ratio of the Simple Strategy

This table reports the monthly CAPM-alphas (α^{CAPM}), monthly Fama-French 5-factor-alphas (α^{FF5}), and annualized Sharpe ratios (SR) of the buy-and-hold and simple strategies investing in the ten beta-sorted portfolios. We present the results for the various aggregation methods used to compute the expected market return and for the forecasting horizon $h = 3$. The results for the forecasting horizons $h = 1, 6, 12$ are presented in the Internet Appendix. The alphas in italics are statistically significant at the 5% level. The sample period is from July 1977 to December 2022.

				Aggregation method for the expected market return								
Buy-and-hold				Median			Mean			PCA		
α^{CAPM}	α^{FF5}	SR		α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR
(1)	(2)	(3)		(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.0007	<i>-0.0023</i>	0.3868	0.0007	-0.0012	0.3654	0.0018	0.0005	0.4760	0.0004	-0.0010	0.3145
2	0.0010	-0.0010	0.4531	<i>0.0032</i>	0.0014	0.6909	<i>0.0042</i>	<i>0.0028</i>	0.8189	<i>0.0024</i>	0.0009	0.6042
3	<i>0.0012</i>	-0.0003	0.4676	<i>0.0040</i>	<i>0.0020</i>	0.7828	<i>0.0046</i>	<i>0.0032</i>	0.8624	<i>0.0032</i>	0.0017	0.7050
4	<i>0.0027</i>	0.0010	0.5715	<i>0.0053</i>	<i>0.0035</i>	0.9202	<i>0.0060</i>	<i>0.0046</i>	0.9963	<i>0.0041</i>	<i>0.0026</i>	0.8012
5	0.0008	-0.0004	0.4411	<i>0.0047</i>	<i>0.0030</i>	0.8226	<i>0.0054</i>	<i>0.0041</i>	0.8868	<i>0.0038</i>	<i>0.0025</i>	0.7283
6	0.0007	<i>-0.0012</i>	0.4351	<i>0.0037</i>	<i>0.0024</i>	0.7111	<i>0.0049</i>	<i>0.0041</i>	0.8184	<i>0.0032</i>	<i>0.0024</i>	0.6605
7	-0.0004	-0.0001	0.3790	<i>0.0044</i>	<i>0.0040</i>	0.7297	<i>0.0057</i>	<i>0.0055</i>	0.8338	<i>0.0046</i>	<i>0.0044</i>	0.7413
8	-0.0012	-0.0005	0.3405	<i>0.0045</i>	<i>0.0044</i>	0.7202	<i>0.0057</i>	<i>0.0058</i>	0.8057	<i>0.0043</i>	<i>0.0042</i>	0.6958
9	-0.0018	0.0002	0.3144	<i>0.0041</i>	<i>0.0045</i>	0.6172	<i>0.0059</i>	<i>0.0065</i>	0.7190	<i>0.0045</i>	<i>0.0049</i>	0.6322
High β	<i>-0.0035</i>	0.0002	0.2609	0.0039	<i>0.0065</i>	0.5270	<i>0.0053</i>	<i>0.0082</i>	0.5871	0.0039	<i>0.0066</i>	0.5246

The first three columns of Table 4 show that low-beta assets have both large alphas and large Sharpe ratios, whereas high-beta assets feature negative alphas and low Sharpe ratios (Frazzini and Pedersen, 2014). Comparing Columns (1)–(2) to Columns (4)–(5), (7)–(8), and (10)–(11) shows that the simple trading strategy yields substantially higher alphas than those obtained by buying and holding. The majority of the simple strategies have monthly CAPM- alphas and FF5- alphas larger than 0.4%, which corresponds to an annualized risk-adjusted return of 4.8%. The largest differences between the alphas of the simple strategies and those of the buy-and-hold strategies are observed among the high-beta deciles. For example, the CAPM-alpha (FF5-alpha) obtained by buying and holding the ninth-decile portfolio is -0.18% (0.02%), whereas that obtained by implementing the simple strategy investing in the ninth-decile portfolio ranges between 0.41% (0.45%) and 0.59% (0.65%) depending on the aggregation method used to compute the expected market return. Most of the simple strategies' alphas are statistically significant at the 5% level, irrespective of whether the benchmark model is the CAPM or the Fama-French 5-factor model. Furthermore, the Sharpe ratios of the simple strategies are substantially larger than those obtained by buying and holding. The Sharpe ratios of the simple strategies investing in high-beta portfolios are particularly large relative to their buy-and-hold counterparts. Indeed, these are about twice as large as those obtained by buying and holding. To summarize, implementing a simple strategy that exploits the positive beta-return relation observed in high expected market return states yields high risk-adjusted returns and Sharpe ratios, irrespective of the riskiness of the asset in which it invests.

4.2. Mean-Variance Strategy

To provide further evidence that predicting asset returns out-of-sample using the ICAPM is valuable to investors, we consider the optimization problem faced by a mean-variance investor who has a single-period horizon, h , and invests in both a risky asset and a risk-free asset. The dynamic problem is

$$\max_{\{\omega_t\}} \mathbb{E}_t(r_{p,t+h}) - \frac{\gamma}{2} \text{Var}_t(r_{p,t+h}), \quad (10)$$

where γ is the risk aversion coefficient, ω_t is the fraction of wealth invested in the risky asset at time t , $r_{p,t+h} = r_{ft} + \omega_t r_{t+h}$ is the investor's portfolio return between time t and $t + h$, r_{ft} is the risk-free rate between time t and $t + h$, and r_{t+h} is the risky asset excess return between time t and $t + h$. The first order condition yields the following optimal fraction of wealth invested in the risky asset at time t

$$\omega_t = \frac{\mathbb{E}_t(r_{t+h})}{\gamma \text{Var}_t(r_{t+h})}.$$

As in [Campbell and Thompson \(2008\)](#), [Rapach et al. \(2010\)](#), [Huang et al. \(2015\)](#), and [Rapach et al. \(2016\)](#) among others, the conditional excess return variance is estimated using a 5-year rolling window of past excess returns. The coefficient of relative risk aversion is set to $\gamma = 3$. The excess return forecast is estimated using the ICAPM, as described in Equation (9). To prevent short selling and leveraged positions when implementing the strategy, we follow [Campbell and Thompson \(2008\)](#), [Rapach et al. \(2010\)](#), and [Neely, Rapach, Tu, and Zhou \(2014\)](#) among others and constrain the fraction of wealth invested in the risky asset to

lie between 0 and 1. That is,

$$\omega_t^* \equiv \max(\min(\omega_t, 1), 0) = \max\left(\min\left(\frac{\mathbb{E}_t(r_{t+h})}{\gamma \text{Var}_t(r_{t+h})}, 1\right), 0\right).$$

As shown in [Cochrane \(1999\)](#) and derived in Internet Appendix IA.1, the maximum Sharpe ratio that a mean-variance investor solving problem (10) can attain by exploiting return predictability is

$$SR^{max} = \sqrt{\frac{(SR_*)^2 + R^2}{(1 - R^2)}}, \quad (11)$$

where R^2 is the R -squared obtained by forecasting the risky asset return, and SR_* is the (non-annualized) Sharpe ratio of the risky asset. Equation (11) shows that the maximum Sharpe ratio of the mean-variance strategy increases with the predictive power of the risky asset return forecast. Given that the risky asset return forecast described in Equation (9) has strong and significant out-of-sample predictive power (see Table 3), the realized Sharpe ratio of the mean-variance strategy is expected to be larger than that of the buy-and-hold strategy.

Table 5
Alpha and Sharpe Ratio of the Mean-Variance Strategy

This table reports the monthly CAPM-alphas (α^{CAPM}), monthly Fama-French 5-factor-alphas (α^{FF5}), and annualized Sharpe ratios (SR) of the buy-and-hold and mean-variance strategies investing in the ten beta-sorted portfolios. We present the results for the various aggregation methods used to compute the expected market return and for the forecasting horizon $h = 3$. The results for $h = 1, 6, 12$ are presented in the Internet Appendix. The alphas in italics are statistically significant at the 5% level. The sample period is from July 1977 to December 2022.

				Aggregation method for the expected market return								
Buy-and-hold				Median			Mean			PCA		
α^{CAPM}	α^{FF5}	SR		α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR
(1)	(2)	(3)		(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.0007	<i>-0.0023</i>	0.3868	0.0005	-0.0011	0.3699	0.0006	-0.0009	0.3740	0.0000	-0.0014	0.3065
2	0.0010	-0.0010	0.4531	<i>0.0030</i>	0.0014	0.6944	<i>0.0030</i>	0.0016	0.7186	<i>0.0024</i>	0.0009	0.6281
3	<i>0.0012</i>	-0.0003	0.4676	<i>0.0032</i>	0.0013	0.7341	<i>0.0033</i>	0.0015	0.7490	<i>0.0035</i>	<i>0.0017</i>	0.7658
4	<i>0.0027</i>	0.0010	0.5715	<i>0.0041</i>	<i>0.0023</i>	0.8214	<i>0.0043</i>	<i>0.0026</i>	0.8460	<i>0.0044</i>	<i>0.0027</i>	0.8593
5	0.0008	-0.0004	0.4411	<i>0.0037</i>	<i>0.0024</i>	0.7682	<i>0.0037</i>	<i>0.0026</i>	0.7788	<i>0.0037</i>	<i>0.0025</i>	0.7629
6	0.0007	<i>-0.0012</i>	0.4351	<i>0.0031</i>	<i>0.0023</i>	0.6998	<i>0.0034</i>	<i>0.0028</i>	0.7309	<i>0.0028</i>	<i>0.0021</i>	0.6641
7	-0.0004	-0.0001	0.3790	<i>0.0031</i>	<i>0.0033</i>	0.6827	<i>0.0036</i>	<i>0.0038</i>	0.7369	<i>0.0038</i>	<i>0.0037</i>	0.7304
8	-0.0012	-0.0005	0.3405	<i>0.0026</i>	<i>0.0033</i>	0.6320	<i>0.0032</i>	<i>0.0038</i>	0.6874	<i>0.0033</i>	<i>0.0036</i>	0.6704
9	-0.0018	0.0002	0.3144	0.0021	<i>0.0031</i>	0.5399	<i>0.0031</i>	<i>0.0040</i>	0.6293	<i>0.0035</i>	<i>0.0042</i>	0.6324
High β	<i>-0.0035</i>	0.0002	0.2609	0.0014	<i>0.0028</i>	0.4364	0.0020	<i>0.0036</i>	0.4781	0.0028	<i>0.0049</i>	0.5251

Table 5 reports the annualized realized Sharpe ratio of the buy-and-hold and mean-variance strategies as well as their monthly CAPM-alphas and monthly Fama-French 5-factor-alphas (FF5-alphas), using the 3-month forecasting horizon. The results obtained using the 1-month, 6-month, and 12-month forecasting horizons are similar and are reported in Table IA2 of the Internet Appendix. As expected, the Sharpe ratios of the mean-variance strategies are substantially larger than those of the buy-and-hold strategies because the portfolio return forecast in Equation (9) has positive predictive power. The difference in Sharpe ratios is particularly pronounced for the strategies investing in high-beta portfolios. Indeed, Sharpe ratios of mean-variance strategies investing in high-beta portfolios are at least 1.5 times larger than those obtained by buying and holding. Furthermore, the mean-variance strategies feature CAPM-alphas and FF5-alphas that are larger than their buy-and-hold counterparts. The majority of the mean-variance strategies' monthly CAPM-alphas and FF5-alphas lie between 0.2% and 0.4%, corresponding to annualized values ranging from 2.4% to 4.8%. Finally, the majority of the mean-variance strategies' alphas are positive and statistically significant at the 5% level, whereas only two out of twenty buy-and-hold alphas are positive and statistically significant at the 5% level.

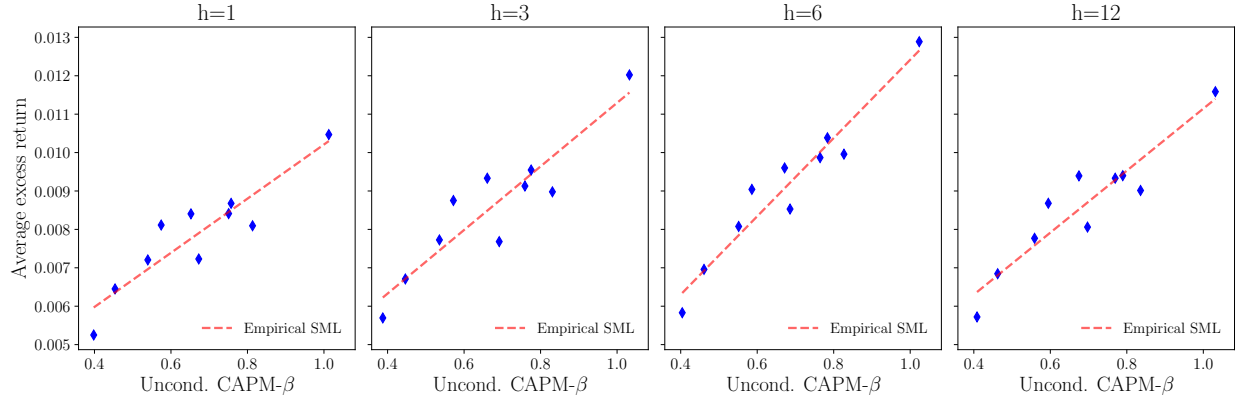
4.3. Trading Strategy with Individual Stocks

This section provides evidence that the positive beta-return relation presented in Figure 3 for trading strategies investing in ten beta-sorted portfolios extends to trading strategies investing in individual stocks. We select each year the largest 500 U.S. stocks based on their market capitalization at the end of the prior year. This choice insures that the results presented in this section are not driven by the liquidity premium or size premium.

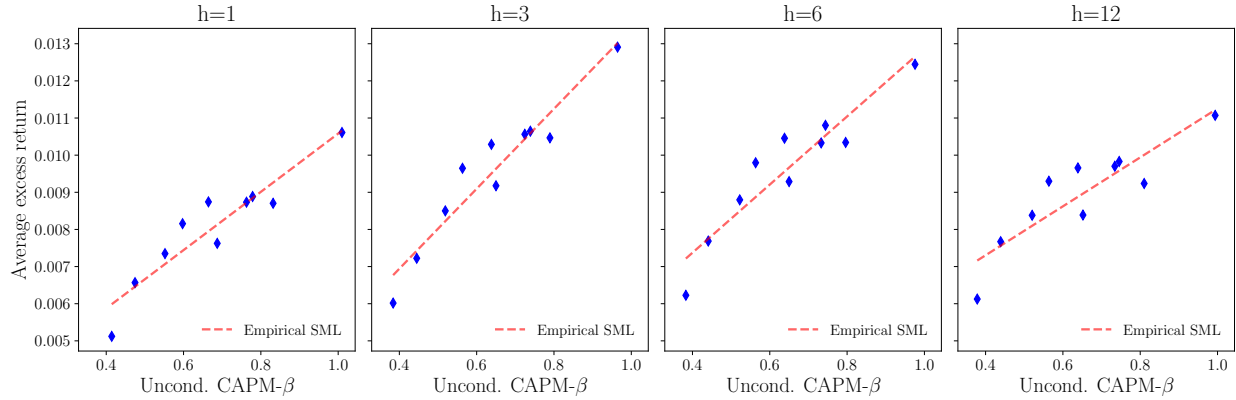
Figure 4. Security Market Line for the Simple Strategies Exploiting the Return Predictability of the Largest 500 U.S. Stocks

This figure presents the security market line for trading strategies that invest 100% of wealth in stock i when the expected market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is above its historical expanding median, and 100% of wealth in the risk-free asset otherwise. The security market line is plotted by beta decile, where the deciles are constructed at every month t using the lagged stock betas, β_{t-1} . $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected excess market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods in Panels A to C, respectively. We select each year the largest 500 U.S. stocks based on their market capitalization at the end of the prior year. The sample period is from July 1977 to December 2022.

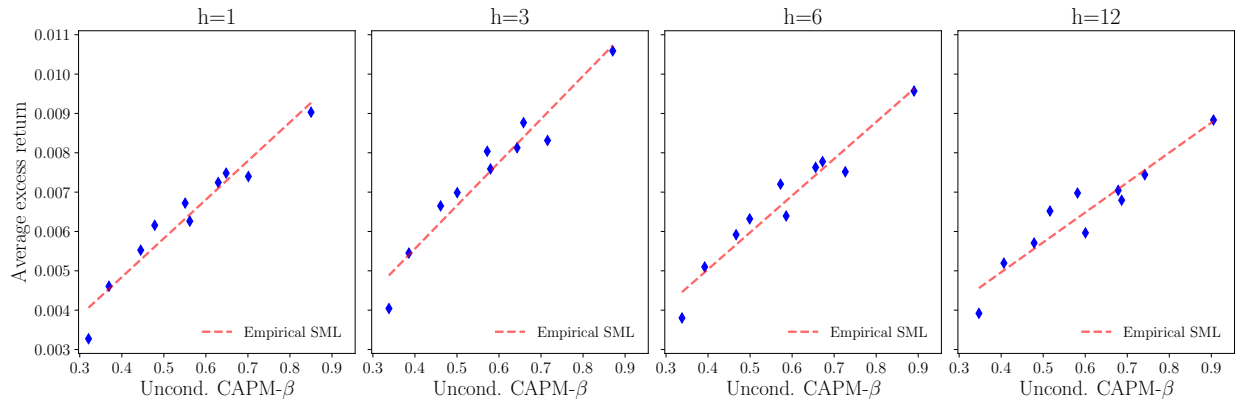
Panel A: Median aggregation method for the expected market return



Panel B: Mean aggregation method for the expected market return



Panel C: PCA aggregation method for the expected market return



Following Section 4.1, the simple trading strategy invests 100% of wealth in stock i if the non-negative expected market return, $\mathbb{E}_t(r_{M,t+h}) \geq 0$, defined in Section 3.2 is above its historical (expanding) median, and long 100% of wealth in the risk-free asset otherwise. Figure 4 plots the security market line of the simple strategies by beta decile, where the deciles are constructed at every month t using the lagged stock betas, β_{t-1} . The security market lines are obtained using different forecasting horizons h and different aggregation methods for the expected market return. Panels A to C report the results obtained using the median, mean, and PCA aggregation methods, respectively. As it was the case for the simple strategy investing in a beta-sorted portfolio (see Figure 3), the average excess return of the simple strategy investing in an individual stock increases with the beta of the underlying stock (see Figure 4). This positive risk-return tradeoff is implied by the positive beta-return relation observed among individual stocks in high expected market return states. Since the simple strategies invest in the stock when the next-period market return is expected to be high and exits this risky position when the next-period market return is expected to be low, the risk-return relation of the simple strategies is positive. To summarize, the simple trading strategies investing in individual stocks feature a positive risk-return tradeoff, as it was the case in Section 4.1 for those investing in ten beta-sorted portfolios.

5. Robustness Checks

We discuss a range of robustness checks for the results provided in Sections 3.3 and 3.4. For brevity, the results discussed in this section refer to tables and figures provided in the Internet Appendix.

5.1. Robustness of the Conditional Beta-Return Relation

This section shows that the conditional beta-return relation reported in Table 2 is robust to alternative sample periods, alternative estimation windows for beta and the market return predictive regressions, alternative high-low cutoff values for the expected market return, and when international portfolios are used as test assets.

5.1.1. Alternative Sample Periods

Table IA3 mirrors Table 2, analyzing the beta-return relationship across three subsamples: 1977–1992, 1993–2007, and 2008–2022. In the first two periods, similar to Table 2, the beta-return relation is significantly weaker in low expected market return states than in high expected market return states across all forecasting horizons and aggregation methods used to compute the expected market return. Over the 2008–2022 sample period, the beta-return relation is significantly weaker in low expected market return states at all horizons for the PCA aggregation method. As in Table 2, the beta-return relation is positive and statistically significant over the 1993–2007 and 2008–2022 sample periods for most forecasting horizons and aggregation methods. Over the 1977–1992 sample period, the beta-return relation is positive and statistically significant at the 1-month and 3-month forecasting horizon for the mean aggregation method, and at the 1-month, 3-month, and 6-month forecasting horizon for the PCA aggregation method. Overall, the conditional beta-return relation computed in Table 2 over the entire 1977–2022 sample period holds over different subsample periods.

5.1.2. Alternative Estimation Windows for Beta

Table IA4 reexamines the results from Table 2 using longer beta estimation windows of 48, 60, and 120 months, instead of 24 months. The results indicate a consistent positive and

statistically significant beta-return relation in high expected market return states across all estimation windows, forecasting horizons, and aggregation methods. The beta-return relation is significantly weaker in low expected market return states than in high expected market return states, showing the robustness of the conditional beta-return relation to varying beta estimation windows.

5.1.3. Alternative High-Low Cutoff Value for the Expected Market Return

We next establish the robustness of results in Table 2 regarding the beta-return relation, considering various high-low cutoff values for expected market returns. Alternative cutoffs include the trailing median, as well as 48, 60, and 120-month rolling medians of realized market returns. Across these cutoffs, Table IA5 reaffirms a positive beta-return relation in high expected market return states, and a substantially weaker relation in low states.

We also consider a conditional high-low cutoff value for the expected market return that depends on inflation. The conditional relation between the expected market return and inflation is obtained by regressing the historical expected market return on the historical inflation rate using an expanding window. The fitted value of the current inflation rate represents the conditional cutoff value (i.e., this value is what the expected market return is expected to be conditional on the current inflation rate). If the current expected market return is above the conditional cutoff value, then the value is considered high, otherwise it is considered low. Consistent with [Fama and Schwert \(1977\)](#), a high inflation rate predicts a low expected market return. Therefore, to be considered high using the inflation-based conditional cutoff value, the current expected market return needs to be much lower in high inflation states than it needs to be in low inflation states. Table IA6 highlights the beta-return relation obtained in high versus low expected market return states, where the high

versus low values are determined using the inflation-based conditional cutoff value. As in Table 2, Table IA6 shows that the beta-return relation is positive and statistically significant in high expected market return states, and significantly weaker in low expected market return states.

In summary, these results underscores the consistency and robustness of the beta-return relation across various definitions of the high-low cutoff value.

5.1.4. Alternative Estimation Windows for the Market Return Predictive Regressions

Table IA7 revisits Table 2's analysis but employs 48, 60, and 120-month rolling windows for estimating individual market return forecasts, replacing the initial expanding window. For a 48-month window, a positive and significant beta-return relation in high expected market return states and a weaker one in low states is observed for most forecasting horizons and aggregation methods. This pattern is also observed with a 60-month window for all forecasting horizons using mean aggregation. With a 120-month window this pattern holds for all forecasting horizons using PCA aggregation and for the 6- and 12-month horizons using mean aggregation. While these findings affirm the robustness of the results to different estimation windows, the expanding window provides more stable estimates and hence a stronger beta-return relation than the rolling windows.

5.1.5. International Portfolios

Table IA8 mirrors Table 2 by focusing on three international Fama-French 25 size-and-book-to-market portfolios sets: Asian (excluding Japan), developed (excluding the U.S.), and European markets. Across these markets, we find a significantly weaker beta-return relation

in low compared to high expected market return states across all forecasting horizons and aggregation methods. In Asian markets, the beta-return relation is positive and significant in high expected return states for all horizons and aggregation methods. In developed markets, the beta-return relation is positive and significant in high expected market return states at the 3-month forecasting horizon for both the mean and PCA aggregation methods, as well as at the 1-month horizon for the mean aggregation. In European markets, the beta-return relation is positive and significant in high expected market return states at the 1-month and 3-month forecasting horizons for the mean aggregation method.

5.2. Robustness of the Portfolio Return Predictability

This section shows that the out-of-sample predictability for the ten beta-sorted portfolios reported in Table 3 is robust to alternative estimation windows for beta, the intertemporal hedging component, and the market return predictive regressions. The results are also robust when the intertemporal hedging component is set to zero, which corresponds to the conditional CAPM specification.

5.2.1. Alternative Estimation Windows for Beta

Table IA9 mirrors the results reported in Table 3 but assuming beta is estimated over a 48-, 60-, and 120-month rolling window instead of a 24-month rolling window. Table IA9 confirms the results reported in Table 3. The OOS- R^2 for the ten beta-sorted portfolios are largely positive and statistically significant, and the estimation window used to estimate beta has a minor effect on the level of the OOS- R^2 . This shows that the out-of-sample predictability of the ten beta-sorted portfolios is robust to alternative rolling estimation windows of beta.

5.2.2. Alternative Estimation Windows for the Intertemporal Hedging Component

Table IA10 is the equivalent of Table 3 but assuming the intertemporal hedging component is estimated over a 48-, 60-, and 120-month rolling window instead of an expanding window. Table IA10 shows that, at the 1-month forecasting horizon, most OOS- R^2 are negative and not statistically significant. When the expected market return is computed using the mean and PCA aggregation methods, however, most OOS- R^2 obtained at the 3-month and 6-month forecasting horizons are positive, large, and statistically significant, irrespective of the estimation window used to estimate the intertemporal hedging component. At the 12-month forecasting horizon, most OOS- R^2 are positive and statistically significant for the PCA aggregation method with both the 48-month and 120-month rolling estimations of the intertemporal hedging component. Overall, the results reported in Table IA10 show that estimating the intertemporal hedging component over 48-, 60-, or 120-month rolling windows instead of over an expanding window yields lower OOS- R^2 . Yet, at the 3-month and 6-month forecasting horizons, out-of-sample predictability is still strong irrespective of the window used to estimate the intertemporal hedging component.

5.2.3. Alternative Estimation Windows for the Market Return Predictive Regressions

Table IA11 is the equivalent of Table 3 but assuming that the individual market return forecasts described in Equations (3) and (4) are estimated using a 48-, 60-, and 120-month rolling window instead of over an expanding window. At the 1-month forecasting horizon, most OOS- R^2 are negative and not statistically significant. At the 3-month, 6-month, and 12-month horizons, however, most OOS- R^2 are positive, large, and statistically significant,

irrespective of both the aggregation method used to compute the expected market return and rolling window used to estimate the individual market return forecasts. Overall, the results reported in Table IA11 show that the out-of-sample return predictability of the beta-sorted portfolios is largely robust to estimating the individual market return forecasts over rolling windows instead of over an expanding window. Yet, the expanding window estimation yields stronger out-of-sample predictability.

5.2.4. Return Forecast Based on the Conditional CAPM (i.e., $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$)

Table IA12 is the equivalent of Table 3 but assuming portfolio i return forecast is computed using the conditional CAPM specification, $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$, instead of the ICAPM specification (9), $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h}) + c_{i,t+h}$. As in Table 3, most OOS- R^2 reported in Table IA12 are positive and statistically significant. This shows that beta-sorted portfolio returns are not only predictable when using the ICAPM specification as forecasting model, but also when using the conditional CAPM specification.

6. Implications for the CAPM Performance over Specific Periods

There is evidence of a positive beta-return relation over specific subsample periods such as overnight (Hendershott et al., 2020), on macroeconomic announcement days (Savor and Wilson, 2014), on FOMC announcement days (Ben-Rephael et al., 2021), and in low inflation months (Cohen et al., 2005). In this section, we show that all of these subsample periods feature particularly high market returns. This together with the positive beta-return relation uncovered in high market return states (see Section 3.3) suggests that the positive beta-return

relation found over the aforementioned subsample periods is likely driven by the high level of market returns observed during these periods.

The empirical findings of [Hendershott et al. \(2020\)](#), [Savor and Wilson \(2014\)](#), [Ben-Rephael et al. \(2021\)](#), and [Cohen et al. \(2005\)](#) show that the beta-return relation (security market line) is positive overnight, on macroeconomic announcement days, on FOMC announcement days, and in months of low inflation, respectively. These findings are confirmed by the Fama-MacBeth and pooled regression results reported in Panels A to D of Table 6.¹⁰ The beta-return relation is positive and statistically significant over these subsample periods. As shown in the first column of the panels, these subsample periods feature particularly high market returns, substantially higher than over the rest of the sample. Indeed, the average market return is as high as 4 bps overnight, 6 bps on macro announcement days, 29 bps on FOMC days, and 1.1% on low inflation months, whereas it is only 0 bps during regular trading hours, 2 bps on non-macro announcement days, 3 bps on non-FOMC days, and 0.2% on high inflation months. These results therefore suggest that the positive beta-return relation uncovered overnight, on macroeconomic announcement days, on FOMC announcement days, and in low inflation periods might be driven by an important feature common to all of these subsample periods: particularly high market returns.

¹⁰The test assets in Panels A to C and Panel D consist of 10-daily- and 10-monthly-beta-sorted portfolios, respectively. For the daily portfolios, we sort on stocks CAPM-beta estimated from a rolling regression of the past 250-day stock excess returns on excess market returns.

Table 6
Asset Returns vs. Beta in Different Time Periods

The table reports estimates from Fama-MacBeth regressions and pooled regressions of excess returns on betas for ten beta-sorted portfolios. Estimates for Fama-MacBeth regressions are computed separately for night and day returns, days with and without macroeconomic announcements, days with and without FOMC announcements, and for months with low and high inflation, in Panels A to D, respectively. β corresponds to a lag rolling 250-day and 24-month portfolio CAPM-beta in Panels A to C and Panel D, respectively. We use night (close-to-open) and day (open-to-close) returns for Panel A as computed in [Hendershott et al. \(2020\)](#), daily (close-to-close) returns for Panels B and C and monthly returns for Panel D. Macroeconomic announcements in Panel B refer to inflation, unemployment, and FOMC announcements. High and low inflation in Panel D corresponds to months with inflation above and below historical median inflation since 1930. Mean r_M corresponds to the average market return computed using the SPDR S&P 500 Trust ETF in Panel A, the average daily excess market returns in Panels B and C, and the average monthly excess market returns in Panel D. Standard errors are reported in parentheses. For Fama-MacBeth regressions, they are calculated using the standard deviation of the time-series coefficient estimates. For pooled regressions, they are calculated using clustered standard errors by date. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The sample periods are specified in the panels' headers.

Panel A. Day and night (1993-2022)									
	Mean r_M	Fama-MacBeth regression				Pooled regression			
		Intercept	β	Avg. R^2	N	Intercept	β	$\beta \times \mathbb{1}_{[Night]}$	$\mathbb{1}_{[Night]}$
Night	0.0004	-0.0006*** (0.0001)	0.0010*** (0.0001)	0.12	7532	0.0008*** (0.0002)	-0.0007*** (0.0002)	0.0015*** (0.0002)	-0.0013*** (0.0002)
Day	0.0000	0.0008*** (0.0001)	-0.0006*** (0.0002)	0.12	7532				
Diff.	0.0004	-0.0014*** (0.000)	0.0017*** (0.000)						
Panel B. Macro announcements (1964-2022)									
	Mean r_M	Fama-MacBeth regression				Pooled regression			
		Intercept	β	Avg. R^2	N	Intercept	β	$\beta \times \mathbb{1}_{[Ann]}$	$\mathbb{1}_{[Ann]}$
Ann	0.0006	-0.0004 (0.0003)	0.0010** (0.0004)	0.33	1069	0.0003** (0.0001)	-0.0001 (0.0002)	0.0008 (0.0005)	-0.0004 (0.0004)
No ann	0.0002	0.0003** (0.0001)	-0.0000 (0.0001)	0.34	13784				
Diff.	0.0003	-0.0007** (0.000)	0.0010** (0.000)						
Panel C. FOMC announcements (1994-2022)									
	Mean r_M	Fama-MacBeth regression				Pooled regression			
		Intercept	β	Avg. R^2	N	Intercept	β	$\beta \times \mathbb{1}_{[FOMC]}$	$\mathbb{1}_{[FOMC]}$
FOMC	0.0029	-0.0020** (0.0009)	0.0048*** (0.0014)	0.39	231	0.0003 (0.0002)	-0.0000 (0.0003)	0.0033*** (0.0012)	-0.0006 (0.0009)
Non FOMC	0.0003	0.0002 (0.0002)	0.0001 (0.0002)	0.39	7050				
Diff.	0.0026	-0.0022** (0.001)	0.0047*** (0.001)						
Panel D. Inflation (1930-2022)									
	Mean r_M	Fama-MacBeth regression				Pooled regression			
		Intercept	β	Avg. R^2	N	Intercept	β	$\beta \times \mathbb{1}_{[Low infl]}$	$\mathbb{1}_{[Low infl]}$
Low infl	0.0111	0.0023 (0.0030)	0.0090*** (0.0034)	0.40	555	0.0086*** (0.0028)	-0.0061* (0.0032)	0.0156*** (0.0056)	-0.0064 (0.0045)
High infl	0.0020	0.0064*** (0.0023)	-0.0043 (0.0027)	0.35	554				
Diff.	0.0091	-0.0041 (0.003)	0.0133*** (0.004)						

7. Conclusion

This paper shows that the ICAPM can be used as a powerful predictor of future asset returns. We show that the empirical relation between future asset returns and betas is positive and strong when future market returns are expected to be high, as the ICAPM suggests. In addition, the sum of an intertemporal hedging component and the product of beta and the expected market return predicts asset returns out-of-sample. As a result, timing strategies exploiting the predictive power of the ICAPM have both Sharpe ratios and risk-adjusted returns (alphas) that are substantially larger than those of the corresponding buy-and-hold strategies.

There has been considerable progress in recent years at finding better predictors of market returns (refer to [Rapach and Zhou, 2013](#), for a survey). Our results suggest this remains a fruitful research area, as using better market return predictors within the ICAPM helps generate superior return predictability in the cross-section. Our findings thus provide a simple and practical framework that will benefit investment strategies and portfolio risk management.

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Internet Appendix to

Equity Return Predictability with the ICAPM

Intended for online publication

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IA.1. Derivation of the Maximum Sharpe Ratio

Following [Campbell and Thompson \(2008\)](#), let us assume the following model for the excess return of the risky asset

$$r_{t+1} = \mu + x_t + \epsilon_{t+1},$$

where μ is the unconditional expected excess return, the excess return predictor x_t has mean zero and variance σ_x^2 , ϵ_{t+1} has mean zero and variance σ_ϵ^2 , and x_t is independent of ϵ_{t+1} .

A mean-variance investor who has a single-period horizon and who can invest in a risk-free asset and a risky asset solves the following problem

$$\max_{\{\omega_t\}} \mathbb{E}_t(r_{p,t+1}) - \frac{\gamma}{2} \text{Var}_t(r_{p,t+1}),$$

where γ is the coefficient of relative risk aversion, ω_t is the fraction of wealth invested in the risky asset at time t , $r_{p,t+1} = r_{ft} + \omega_t r_{t+1}$ is the investor's portfolio return between time t and $t + 1$, r_{ft} is the risk-free rate between time t and $t + 1$, and r_{t+1} is the risky asset excess return between time t and $t + 1$. The first order condition yields the following optimal fraction of wealth invested in the risky asset at time t

$$\omega_t = \frac{\mathbb{E}_t(r_{t+1})}{\gamma \text{Var}_t(r_{t+1})} = \frac{\mu + x_t}{\gamma \sigma_\epsilon^2}.$$

Therefore, the unconditional expected excess return of the strategy satisfies

$$\begin{aligned}
\mathbb{E}(r_{p,t+1} - r_{ft}) &= \mathbb{E}(\mathbb{E}_t(\omega_t r_{t+1})) \\
&= \frac{\mathbb{E}((\mu + x_t)^2)}{\gamma \sigma_\epsilon^2} \\
&= \frac{\mu^2 + \sigma_x^2}{\gamma \sigma_\epsilon^2} \\
&= \frac{(SR)^2 + R^2}{\gamma(1 - R^2)},
\end{aligned}$$

where $SR = \frac{\mu}{\sqrt{\sigma_x^2 + \sigma_\epsilon^2}}$ is the Sharpe ratio of the risky asset and $R^2 = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\epsilon^2}$ is the R^2 obtained by regressing the excess return r_{t+1} on the predictor x_t .

The expected conditional excess return variance of the strategy satisfies

$$\begin{aligned}
\mathbb{E}(\text{Var}_t(r_{p,t+1} - r_{ft})) &= \mathbb{E}(\text{Var}_t(\omega_t r_{t+1})) \\
&= \mathbb{E}(\omega_t^2 \sigma_\epsilon^2) \\
&= \frac{\mu^2 + \sigma_x^2}{\gamma^2 \sigma_\epsilon^2} \\
&= \frac{(SR)^2 + R^2}{\gamma^2(1 - R^2)}.
\end{aligned}$$

The true Sharpe ratio, SR^{true} , of the strategy satisfies

$$\begin{aligned}
SR^{true} &= \frac{\mathbb{E}(r_{p,t+1} - r_{ft})}{\sqrt{\text{Var}(r_{p,t+1} - r_{ft})}} = \frac{\mathbb{E}(r_{p,t+1} - r_{ft})}{\sqrt{\mathbb{E}(\text{Var}_t(r_{p,t+1} - r_{ft})) + \text{Var}(\mathbb{E}_t(r_{p,t+1} - r_{ft}))}} \\
&\leq \frac{\mathbb{E}(r_{p,t+1} - r_{ft})}{\sqrt{\mathbb{E}(\text{Var}_t(r_{p,t+1} - r_{ft}))}} \\
&= \sqrt{\frac{(SR)^2 + R^2}{(1 - R^2)}} \equiv SR^{max}.
\end{aligned}$$

As shown in [Cochrane \(1999\)](#), SR^{max} is the maximum Sharpe ratio that a mean-variance

investor can attain by exploiting return predictability.

IA.2. Out-of-Sample Predictability of Market Returns without Restricting the Expected Market Return

Figure IA1 plots the cumulative squared prediction errors of the *null* minus the cumulative squared prediction error of the *alternative*. The *alternative* is the model that relies on a given predictive variable, as described in the legend of the graph. The *null* is a model for which the forecast is the historical mean of market returns. In Figure IA1, the market return forecast is not restricted and can therefore take negative values. The left panels show that the predictive power of investor sentiment spikes up during the dot-com bubble, whereas that of short-interest, variance risk premium, and investor attention spikes up during the financial crisis. During the COVID-19 crisis, the predictive power of the variance risk premium collapses, while that of short-interest increases substantially. The right panels of Figure IA1 shows that aggregating the individual forecasts improves substantially the out-of-sample predictive power. Indeed, the predictive power of the aggregated forecasts increases gradually over time (up to the COVID-19 crisis), whereas that of the individual forecasts remains close to zero and is very volatile. Note that, as described in Section 3.2, the PCA aggregation method assigns time-varying weights to the individual forecasts that take into account the relevance of each individual predictor. As a result, the market return forecast computed with the PCA aggregation method has the largest predictive power when the non-negativity constraint, $\mathbb{E}_t(r_{M,t+h}) \geq 0$, is imposed (see Column (6) of Table 1).

IA.3. Performance of the Simple and Mean-Variance Strategies with forecasting horizon $h \in \{1, 6, 12\}$

Tables IA1 and IA2 are the equivalents of Tables 4 and 5, respectively, for the forecasting horizons $h = 1$ in Panel A, $h = 6$ in Panel B, and $h = 12$ in Panel C. As it is the case at the 3-month forecasting horizon, the alphas and Sharpe ratios of the simple and mean-variance strategies at the 1-month, 6-month, and 12-month horizons are substantially larger than those obtained by buying and holding. Most alphas obtained with the simple and mean-variance strategies are positive and statistically significant, and their Sharpe ratios are about double those obtained by buying and holding.

IA.4. Robustness of the Conditional Beta-Return Relation

This section shows that the conditional beta-return relation reported in Table 2 is robust to alternative sample periods, alternative estimation windows for beta and the market return predictive regressions, alternative high-low cutoff values for the expected market return, and when international portfolios are used as test assets.

IA.4.1. Alternative Sample Periods

Table IA3 provides the results reported in Table 2 but computed over three alternative subsample periods: 1977–1992, 1993–2007, and 2008–2022. As in Table 2, the beta-return relation is significantly weaker in low expected market return states than in high expected market return states over the 1977–1992 and 1993–2007 sample periods, for all forecasting horizons and aggregation methods used to compute the expected market return. Over the 2008–2022 sample period, the beta-return relation is significantly weaker in low expected market return states at all horizons for the PCA aggregation method. As in Table 2, the

beta-return relation is positive and statistically significant over the 1993–2007 and 2008–2022 sample periods for most forecasting horizons and aggregation methods. Over the 1977–1992 sample period, the beta-return relation is positive and statistically significant at the 1-month and 3-month forecasting horizon for the mean aggregation method, and at the 1-month, 3-month, and 6-month forecasting horizon for the PCA aggregation method. Overall, the conditional beta-return relation computed in Table 2 over the entire 1977–2022 sample period hold over different subsample periods.

IA.4.2. Alternative Estimation Windows for Beta

Table IA4 is the equivalent of Table 2 but assuming beta is estimated over a 48-, 60-, and 120-month rolling window instead of a 24-month rolling window. Table IA4 shows that the beta-return relation is positive and statistically significant when market returns are expected to be high for all estimation windows, forecasting horizons, and aggregation methods used to compute the expected market return. In addition, the beta-return relation is significantly weaker when market returns are expected to be low, irrespective of the CAPM- β estimation windows, forecasting horizons, and aggregation methods. These results show that the conditional beta-return relation is robust to alternative rolling estimation windows for beta.

IA.4.3. Alternative High-Low Cutoff Value for the Expected Market Return

To show that the results presented in Table 2 are robust to several definitions of the high-low cutoff value for the expected market return, one considers here the following alternative high-low cutoff values:

1. Trailing median of realized market returns

2. 48-month rolling median of realized market returns
3. 60-month rolling median of realized market returns
4. 120-month rolling median of realized market returns

Using these alternative cutoff values, Table IA5 confirms the result presented in Table 2 that the beta-return relation is positive in high expected market return states, and significantly weaker in low expected market return states.

We also consider a conditional high-low cutoff value for the expected market return that depends on inflation by proceeding as follows. The conditional relation between the expected market return and inflation is obtained by regressing the historical expected market return on the historical inflation rate using an expanding window. The fitted value of the current inflation rate represents the conditional cutoff value (i.e., this value is what the expected market return is expected to be conditional on the current inflation rate). If the current expected market return is above the conditional cutoff value, then the value is considered high, otherwise it is considered low. Consistent with [Fama and Schwert \(1977\)](#), a high inflation rate predicts a low expected market return. Therefore, to be considered high using the inflation-based conditional cutoff value, the current expected market return needs to be much lower in high inflation states than it needs to be in low inflation states. Table IA6 highlights the beta-return relation obtained in high versus low expected market return states, where the high versus low values are determined using the inflation-based conditional cutoff value. As in Table 2, Table IA6 shows that the beta-return relation is positive and statistically significant in high expected market return states, and significantly weaker in low expected market return states. To summarize, the beta-return relation is robust to different specifications of the high-low cutoff value.

IA.4.4. Alternative Estimation Windows for the Market Return Predictive Regressions

Table IA7 is the equivalent of Table 2 but assuming that the individual market return forecasts described in Equations (3) and (4) are estimated over a 48-, 60-, and 120-month rolling window instead of an expanding window. When individual market return forecasts are estimated over a 48-month rolling window, the beta-return relation is positive and statistically significant in high expected market return states and significantly weaker in low expected market return states for the majority of the forecasting horizons and aggregation methods used to compute the expected market return. When individual market return forecasts are estimated over a 60-month rolling window, the aforementioned conditional beta-return relation holds at all horizons for the mean aggregation method. When individual market return forecasts are estimated over a 120-month rolling window, the aforementioned conditional beta-return relation holds at all horizons for the PCA aggregation method and at the 6-month and 12-month horizons for the mean aggregation method. Overall, this section shows that the results reported in Table 2 are largely robust to estimating individual market return forecasts using rolling windows instead of an expanding window. However, estimating individual market return forecasts using an expanding window provides more stable estimates than the rolling window estimates. As a result, the beta-return relation obtained with the expanding window predictive regression is clearer and stronger than those obtained with the rolling window predictive regressions.

IA.4.5. International Portfolios

Table IA8 is the equivalent of Table 2 but using three sets of international portfolios as test assets: the Fama-French 25 size-and-book-to-market portfolios in Asian markets excluding

Japan, developed markets excluding the U.S., and European markets. As in Table 2, the beta-return relation observed in Asian markets, developed markets, and European markets is significantly weaker in low expected market return states than in high expected market return states. This result holds for all forecasting horizons and aggregation methods used to compute the expected market return. The beta-return relation observed in Asian markets is positive and statistically significant in high expected market return states at all forecasting horizons and for all aggregation methods used to compute the expected market return. In developed markets, the beta-return relation in high expected market return states is positive and statistically significant at the 3-month forecasting horizon for both the mean and PCA aggregation methods, as well as at the 1-month forecasting horizon for the mean aggregation method. In European markets, the beta-return relation in high expected market return states is positive and statistically significant at the 1-month and 3-month forecasting horizons for the mean aggregation method.

IA.5. Robustness of the Portfolio Return Predictability

This section shows that the out-of-sample predictability for the ten beta-sorted portfolios reported in Table 3 is robust to alternative estimation windows for beta, the intertemporal hedging component, and the market return predictive regressions. The results are also robust when the intertemporal hedging component is set to zero, which corresponds to the conditional CAPM specification.

IA.5.1. Alternative Estimation Windows for Beta

Table IA9 mirrors the results reported in Table 3 but assuming beta is estimated over a 48-, 60-, and 120-month rolling window instead of a 24-month rolling window. Table IA9 confirms

the results reported in Table 3. The OOS- R^2 for the ten beta-sorted portfolios are largely positive and statistically significant, and the estimation window used to estimate beta has a minor effect on the level of the OOS- R^2 . This shows that the out-of-sample predictability of the ten beta-sorted portfolios is robust to alternative rolling estimation windows of beta.

IA.5.2. Alternative Estimation Windows for the Intertemporal Hedging Component

Table IA10 is the equivalent of Table 3 but assuming the intertemporal hedging component is estimated over a 48-, 60-, and 120-month rolling window instead of an expanding window. Table IA10 shows that, at the 1-month forecasting horizon, most OOS- R^2 are negative and not statistically significant. When the expected market return is computed using the mean and PCA aggregation methods, however, most OOS- R^2 obtained at the 3-month and 6-month forecasting horizons are positive, large, and statistically significant, irrespective of the estimation window used to estimate the intertemporal hedging component. At the 12-month forecasting horizon, most OOS- R^2 are positive and statistically significant for the PCA aggregation method with both the 48-month and 120-month rolling estimations of the intertemporal hedging component. Overall, the results reported in Table IA10 show that estimating the intertemporal hedging component over 48-, 60-, or 120-month rolling windows instead of over an expanding window yields lower OOS- R^2 . Yet, at the 3-month and 6-month forecasting horizons, out-of-sample predictability is still strong irrespective of the window used to estimate the intertemporal hedging component.

IA.5.3. Alternative Estimation Windows for the Market Return Predictive Regressions

Table IA11 is the equivalent of Table 3 but assuming that the individual market return forecasts described in Equations (3) and (4) are estimated using a 48-, 60-, and 120-month rolling window instead of an expanding window. At the 1-month forecasting horizon, most OOS- R^2 are negative and not statistically significant. At the 3-month, 6-month, and 12-month horizons, however, most OOS- R^2 are positive, large, and statistically significant, irrespective of both the aggregation method used to compute the expected market return and rolling window used to estimate the individual market return forecasts. Overall, the results reported in Table IA11 show that the out-of-sample return predictability of the beta-sorted portfolios is largely robust to estimating the individual market return forecasts over rolling windows instead of over an expanding window. Yet, the expanding window estimation yields stronger out-of-sample predictability.

IA.5.4. Return Forecast Based on the Conditional CAPM (i.e., $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$)

Table IA12 is the equivalent of Table 3 but assuming portfolio i return forecast is computed using the conditional CAPM specification, $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h})$, instead of the ICAPM specification (9), $\mathbb{E}_t(r_{i,t+h}) = \beta_{it}\mathbb{E}_t(r_{M,t+h}) + c_{i,t+h}$. As in Table 3, most OOS- R^2 reported in Table IA12 are positive and statistically significant. This shows that beta-sorted portfolio returns are not only predictable when using the ICAPM specification as forecasting model, but also when using the conditional CAPM specification.

IA.6. Sentiment-Based versus Price-Based Forecasts of the Market Return

Table IA13 investigates the impact of considering a price-based forecast of the market return (i.e., aggregation of the individual forecasts obtained with the variance risk premium and short-interest) versus a sentiment-based forecast of the market return (i.e., aggregation of the individual forecasts obtained with investor sentiment, managerial sentiment, and investor attention) on the beta-return relation. Panel A shows that the beta-return relation obtained with the price-based forecast of the market return is positive and statistically significant in high expected market return states, and significantly weaker in low expected market return states, for all forecasting horizons. That is, the results are similar to those reported in Table 2, where the forecasted market return is an aggregation of *all* individual forecasts. Panel B, however, shows that the beta-return relation obtained with the sentiment-based forecast of the market return is positive and statistically significant in high expected market return states, and significantly weaker in low expected market return states, for the 6-month and 12-month forecasting horizons but not for the 1-month and 3-month.

IA.7. Figures and Tables

Figure IA1. Evolution of the Out-of-Sample Predictive Ability of Market Return Forecasts, Assuming no Restrictions on the Forecasts

This figure shows the evolution of the out-of-sample predictive performance of individual market return forecasts (left panels) and aggregated market return forecasts (right panels). Specifically, these are the cumulative squared prediction errors of the *null* minus the cumulative squared prediction error of the *alternative*. The *alternative* is the model that relies on a given predictive variable, as described in the legend of the graph. The *null* is a model for which the forecast is the historical mean of market returns. Each row corresponds to a different forecast horizon h for $h = 1, 3, 6, 12$. The out-of-sample period is from July 1977 to December 2022.

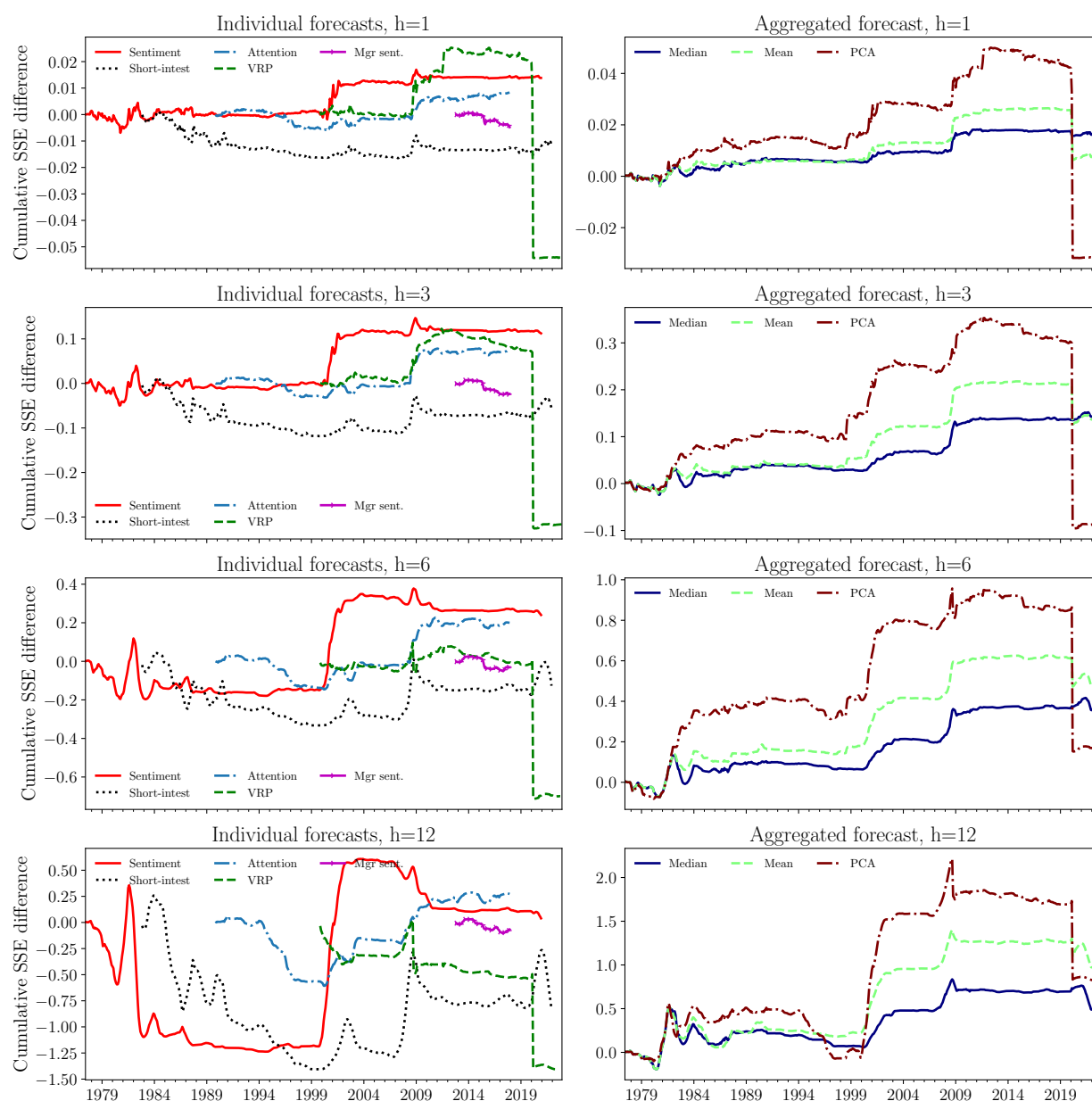


Table IA1
Alpha and Sharpe Ratio of the Simple Strategy

This table reports the monthly CAPM-alphas (α^{CAPM}), monthly Fama-French 5-factor-alphas (α^{FF5}), and annualized Sharpe ratios (SR) of the buy-and-hold and simple strategies investing in the ten beta-sorted portfolios. We present the results for the various expected market return aggregation methods and for $h = 1, 6, 12$ in Panels A to C, respectively. $h = 3$ is reported in the paper. The alphas in italics are statistically significant at the 5% level.

Panel A. $h = 1$												
Aggregation method for the expected market return												
	Buy-and-hold			Median			Mean			PCA		
	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.0007	-0.0023	0.3868	0.0003	-0.0015	0.3285	0.0003	-0.0013	0.3343	0.0000	-0.0014	0.2734
2	0.0010	-0.0010	0.4531	0.0028	0.0011	0.6511	0.0029	0.0012	0.6643	0.0019	0.0004	0.5393
3	0.0012	-0.0003	0.4676	0.0035	0.0017	0.7381	0.0034	0.0017	0.7296	0.0024	0.0008	0.6101
4	0.0027	0.0010	0.5715	0.0047	0.0030	0.8606	0.0046	0.0030	0.8482	0.0033	0.0017	0.7065
5	0.0008	-0.0004	0.4411	0.0039	0.0023	0.7524	0.0040	0.0025	0.7670	0.0029	0.0015	0.6395
6	0.0007	-0.0012	0.4351	0.0027	0.0014	0.6315	0.0031	0.0020	0.6678	0.0024	0.0015	0.5854
7	-0.0004	-0.0001	0.3790	0.0038	0.0031	0.6865	0.0038	0.0032	0.6879	0.0035	0.0031	0.6437
8	-0.0012	-0.0005	0.3405	0.0037	0.0031	0.6623	0.0037	0.0033	0.6614	0.0033	0.0030	0.6111
9	-0.0018	0.0002	0.3144	0.0028	0.0027	0.5408	0.0035	0.0036	0.5817	0.0034	0.0036	0.5528
High β	-0.0035	0.0002	0.2609	0.0025	0.0043	0.4623	0.0025	0.0048	0.4691	0.0031	0.0055	0.4746

Table IA1
Alpha and Sharpe Ratio of the Simple Strategy (Cont.)

Panel B. $h = 6$												
Forecast aggregation method												
	Buy-and-hold			Median			Mean			PCA		
	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.0007	-0.0023	0.3868	0.0008	-0.0010	0.3733	0.0019	0.0007	0.4858	-0.0000	-0.0014	0.2715
2	0.0010	-0.0010	0.4531	0.0034	0.0016	0.7152	0.0047	0.0032	0.8724	0.0021	0.0006	0.5661
3	0.0012	-0.0003	0.4676	0.0042	0.0024	0.8138	0.0050	0.0036	0.9108	0.0027	0.0011	0.6380
4	0.0027	0.0010	0.5715	0.0056	0.0040	0.9541	0.0061	0.0048	1.0113	0.0035	0.0020	0.7386
5	0.0008	-0.0004	0.4411	0.0050	0.0034	0.8548	0.0055	0.0043	0.9005	0.0030	0.0017	0.6516
6	0.0007	-0.0012	0.4351	0.0041	0.0028	0.7464	0.0051	0.0043	0.8373	0.0024	0.0015	0.5879
7	-0.0004	-0.0001	0.3790	0.0053	0.0047	0.8054	0.0058	0.0055	0.8366	0.0035	0.0031	0.6474
8	-0.0012	-0.0005	0.3405	0.0053	0.0049	0.7759	0.0056	0.0055	0.7921	0.0035	0.0033	0.6285
9	-0.0018	0.0002	0.3144	0.0051	0.0051	0.6754	0.0056	0.0061	0.6999	0.0034	0.0036	0.5572
High β	-0.0035	0.0002	0.2609	0.0048	0.0070	0.5699	0.0046	0.0074	0.5594	0.0031	0.0056	0.4747
Panel C. $h = 12$												
Forecast aggregation method												
	Buy-and-hold			Median			Mean			PCA		
	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.0007	-0.0023	0.3868	0.0006	-0.0012	0.3662	0.0016	0.0002	0.4561	-0.0000	-0.0012	0.2805
2	0.0010	-0.0010	0.4531	0.0030	0.0012	0.6773	0.0046	0.0031	0.8640	0.0022	0.0009	0.5922
3	0.0012	-0.0003	0.4676	0.0040	0.0021	0.7831	0.0048	0.0031	0.8748	0.0025	0.0012	0.6358
4	0.0027	0.0010	0.5715	0.0051	0.0033	0.9027	0.0057	0.0041	0.9581	0.0037	0.0024	0.7592
5	0.0008	-0.0004	0.4411	0.0045	0.0028	0.8089	0.0050	0.0036	0.8510	0.0026	0.0014	0.6216
6	0.0007	-0.0012	0.4351	0.0036	0.0022	0.7068	0.0044	0.0035	0.7717	0.0021	0.0012	0.5648
7	-0.0004	-0.0001	0.3790	0.0044	0.0036	0.7362	0.0047	0.0044	0.7546	0.0029	0.0024	0.6043
8	-0.0012	-0.0005	0.3405	0.0043	0.0036	0.7116	0.0048	0.0046	0.7306	0.0029	0.0025	0.5906
9	-0.0018	0.0002	0.3144	0.0038	0.0034	0.6044	0.0039	0.0045	0.5930	0.0028	0.0023	0.5190
High β	-0.0035	0.0002	0.2609	0.0032	0.0048	0.4998	0.0032	0.0060	0.4876	0.0021	0.0036	0.4308

Table IA2
Alpha and Sharpe Ratio of the Mean-Variance Strategy

This table reports the monthly CAPM-alphas (α^{CAPM}), monthly Fama-French 5-factor-alphas (α^{FF5}), and annualized Sharpe ratios (SR) of the buy-and-hold and mean-variance strategies investing in the ten beta-sorted portfolios. We present the results for the various expected market return aggregation methods and for $h = 1, 6, 12$ in Panels A to C, respectively. $h = 3$ is reported in the paper. The alphas in italics are statistically significant at the 5% level.

Panel A. $h = 1$												
Aggregation method for the expected market return												
Buy-and-hold			Median			Mean			PCA			
α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	α^{CAPM}	α^{FF5}	SR	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
Low β	0.0007	-0.0023	0.3868	-0.0001	-0.0018	0.3015	-0.0002	-0.0018	0.2967	-0.0006	-0.0021	0.2405
2	0.0010	-0.0010	0.4531	0.0024	0.0007	0.6216	0.0022	0.0005	0.5954	0.0020	0.0004	0.5771
3	0.0012	-0.0003	0.4676	0.0028	0.0009	0.6864	0.0028	0.0010	0.6866	0.0029	0.0011	0.7021
4	0.0027	0.0010	0.5715	0.0036	0.0019	0.7734	0.0036	0.0019	0.7716	0.0037	0.0020	0.7863
5	0.0008	-0.0004	0.4411	0.0032	0.0018	0.7113	0.0027	0.0015	0.6748	0.0028	0.0016	0.6783
6	0.0007	-0.0012	0.4351	0.0024	0.0016	0.6313	0.0021	0.0015	0.6060	0.0019	0.0011	0.5691
7	-0.0004	-0.0001	0.3790	0.0021	0.0023	0.5954	0.0023	0.0023	0.6106	0.0025	0.0022	0.6115
8	-0.0012	-0.0005	0.3405	0.0015	0.0024	0.5397	0.0021	0.0025	0.5838	0.0020	0.0022	0.5567
9	-0.0018	0.0002	0.3144	0.0007	0.0021	0.4373	0.0016	0.0026	0.5134	0.0019	0.0025	0.5163
High β	-0.0035	0.0002	0.2609	-0.0000	0.0016	0.3182	0.0006	0.0021	0.3658	0.0018	0.0037	0.4468

Table IA2
Alpha and Sharpe Ratio of the Mean-Variance Strategy (Cont.)

Panel B. $h = 6$												
Forecast aggregation method												
	Buy-and-hold			α^{CAPM}	Median		α^{CAPM}	Mean		α^{CAPM}	PCA	
	α^{CAPM}	α^{FF5}	SR		α^{FF5}	SR		α^{FF5}	SR		α^{FF5}	SR
	(1)	(2)	(3)		(5)	(6)		(8)	(9)		(11)	(12)
Low β	0.0007	-0.0023	0.3868	0.0005	-0.0008	0.3534	0.0005	-0.0006	0.3593	0.0003	-0.0009	0.3282
2	0.0010	-0.0010	0.4531	0.0030	0.0017	0.6990	0.0031	0.0020	0.7332	0.0029	0.0017	0.7041
3	0.0012	-0.0003	0.4676	0.0032	0.0017	0.7252	0.0033	0.0018	0.7416	0.0034	0.0020	0.7598
4	0.0027	0.0010	0.5715	0.0039	0.0027	0.8158	0.0041	0.0029	0.8371	0.0043	0.0030	0.8488
5	0.0008	-0.0004	0.4411	0.0037	0.0029	0.7653	0.0038	0.0031	0.7884	0.0038	0.0030	0.7777
6	0.0007	-0.0012	0.4351	0.0032	0.0028	0.7007	0.0034	0.0031	0.7265	0.0032	0.0027	0.6956
7	-0.0004	-0.0001	0.3790	0.0032	0.0040	0.6829	0.0037	0.0043	0.7433	0.0042	0.0042	0.7777
8	-0.0012	-0.0005	0.3405	0.0028	0.0038	0.6423	0.0035	0.0043	0.7051	0.0033	0.0037	0.6789
9	-0.0018	0.0002	0.3144	0.0021	0.0036	0.5391	0.0030	0.0042	0.6167	0.0028	0.0037	0.5955
High β	-0.0035	0.0002	0.2609	0.0014	0.0028	0.4239	0.0021	0.0036	0.4737	0.0022	0.0039	0.4906

Panel C. $h = 12$												
Forecast aggregation method												
	Buy-and-hold			α^{CAPM}	Median		α^{CAPM}	Mean		α^{CAPM}	PCA	
	α^{CAPM}	α^{FF5}	SR		α^{FF5}	SR		α^{FF5}	SR		α^{FF5}	SR
	(1)	(2)	(3)		(5)	(6)		(8)	(9)		(11)	(12)
Low β	0.0007	-0.0023	0.3868	0.0007	-0.0006	0.3737	0.0007	-0.0004	0.3753	0.0004	-0.0006	0.3461
2	0.0010	-0.0010	0.4531	0.0031	0.0019	0.7309	0.0032	0.0021	0.7513	0.0028	0.0017	0.7034
3	0.0012	-0.0003	0.4676	0.0032	0.0017	0.7270	0.0034	0.0019	0.7465	0.0035	0.0020	0.7674
4	0.0027	0.0010	0.5715	0.0040	0.0027	0.8241	0.0042	0.0030	0.8545	0.0044	0.0032	0.8721
5	0.0008	-0.0004	0.4411	0.0038	0.0029	0.7749	0.0041	0.0033	0.8162	0.0040	0.0033	0.8077
6	0.0007	-0.0012	0.4351	0.0034	0.0030	0.7196	0.0038	0.0035	0.7698	0.0035	0.0029	0.7292
7	-0.0004	-0.0001	0.3790	0.0029	0.0037	0.6652	0.0035	0.0041	0.7344	0.0038	0.0037	0.7503
8	-0.0012	-0.0005	0.3405	0.0032	0.0041	0.6834	0.0038	0.0044	0.7412	0.0038	0.0040	0.7338
9	-0.0018	0.0002	0.3144	0.0020	0.0032	0.5279	0.0030	0.0040	0.6189	0.0032	0.0035	0.6196
High β	-0.0035	0.0002	0.2609	0.0013	0.0028	0.4292	0.0021	0.0035	0.4821	0.0023	0.0034	0.4895

Table IA3
Pooled Regression: Different Subsample Periods

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return, $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below its trailing median and zero otherwise, and $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns. Panels A to C present the results for three 15-year subsamples, 1997-1992, 1993-2007, and 2008-2002, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

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Panel A. 1977-1992												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.002 (0.001)	0.008 (0.005)	0.017 (0.012)	-0.005 (0.021)	0.006*** (0.002)	0.015*** (0.005)	0.022 (0.013)	-0.010 (0.019)	0.002* (0.001)	0.011** (0.005)	0.027** (0.012)	0.034 (0.024)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.010*** (0.003)	-0.037*** (0.011)	-0.093*** (0.025)	-0.131*** (0.039)	-0.017*** (0.005)	-0.050*** (0.013)	-0.101*** (0.032)	-0.112*** (0.041)	-0.011*** (0.003)	-0.053*** (0.015)	-0.118*** (0.030)	-0.212*** (0.040)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.004 (0.003)	0.017 (0.011)	0.059** (0.028)	0.081* (0.044)	0.004 (0.005)	0.011 (0.013)	0.041 (0.037)	-0.002 (0.049)	-0.004 (0.003)	0.010 (0.015)	0.042 (0.031)	0.093** (0.045)
Intercept	0.007*** (0.002)	0.021*** (0.006)	0.039*** (0.013)	0.115*** (0.024)	0.006*** (0.001)	0.022*** (0.005)	0.045*** (0.015)	0.152*** (0.021)	0.011*** (0.001)	0.025*** (0.004)	0.046*** (0.013)	0.109*** (0.025)
N	1,870	1,870	1,870	1,870	1,870	1,870	1,870	1,870	1,870	1,870	1,870	1,870
R^2	0.004	0.016	0.031	0.043	0.019	0.050	0.066	0.115	0.023	0.057	0.100	0.147

Table IA3
Pooled Regression: Different Subsample Periods (Cont.)

Panel B. 1993-2007												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.003 (0.002)	0.017*** (0.002)	0.021*** (0.003)	0.030*** (0.009)	0.002 (0.002)	0.012*** (0.004)	0.013 (0.010)	0.034*** (0.011)	0.008*** (0.003)	0.007 (0.006)	0.048*** (0.007)	0.065*** (0.005)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.008*** (0.002)	-0.042*** (0.009)	-0.054*** (0.012)	-0.084*** (0.014)	-0.010*** (0.002)	-0.044*** (0.004)	-0.064*** (0.009)	-0.208*** (0.011)	-0.013*** (0.001)	-0.018*** (0.003)	-0.088*** (0.010)	-0.150*** (0.043)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.003 (0.003)	-0.012* (0.007)	-0.049*** (0.011)	-0.138*** (0.024)	-0.002 (0.003)	-0.008 (0.008)	-0.034* (0.020)	-0.104*** (0.023)	0.010*** (0.003)	-0.000 (0.007)	0.044*** (0.006)	0.025 (0.031)
Intercept	0.006*** (0.002)	0.015*** (0.002)	0.040*** (0.006)	0.088*** (0.015)	0.006*** (0.002)	0.018*** (0.006)	0.041*** (0.013)	0.078*** (0.018)	0.000 (0.004)	0.019** (0.009)	0.009 (0.012)	0.056*** (0.011)
N	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800	1,800
R^2	0.010	0.083	0.138	0.214	0.011	0.074	0.111	0.337	0.005	0.012	0.064	0.130

Panel C. 2008-2022												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.011*** (0.002)	0.024*** (0.006)	0.049*** (0.011)	0.080*** (0.021)	0.009*** (0.003)	0.026*** (0.006)	0.048*** (0.011)	0.063*** (0.020)	0.016*** (0.003)	0.038*** (0.008)	0.060*** (0.014)	0.096*** (0.023)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.023*** (0.003)	-0.004 (0.013)	-0.030 (0.022)	0.006 (0.032)	-0.013 (0.008)	-0.014 (0.016)	-0.025 (0.026)	0.192*** (0.057)	-0.021*** (0.004)	-0.035*** (0.007)	-0.044*** (0.011)	-0.062*** (0.021)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.008** (0.003)	-0.083*** (0.014)	-0.095*** (0.023)	-0.089** (0.035)	-0.007 (0.008)	-0.046*** (0.018)	-0.089*** (0.027)	-0.319*** (0.052)	0.010*** (0.004)	-0.002 (0.006)	0.002 (0.010)	-0.004 (0.025)
Intercept	0.002 (0.003)	0.013 (0.008)	0.024 (0.016)	0.061* (0.031)	0.002 (0.003)	0.009 (0.008)	0.023 (0.016)	0.079*** (0.031)	-0.003 (0.004)	0.003 (0.010)	0.008 (0.018)	0.049 (0.034)
N	1,790	1,770	1,740	1,680	1,790	1,770	1,740	1,680	1,790	1,770	1,740	1,680
R^2	0.041	0.082	0.082	0.030	0.017	0.050	0.067	0.043	0.016	0.043	0.028	0.033

Table IA4
Pooled Regression: Alternative Rolling Estimation of CAPM- β

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return, $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below its trailing median and zero otherwise, and $\beta_{i,t}$ is the CAPM- β estimated using the previous 48, 60, 120 monthly returns in Panels A to C, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. 48-month rolling estimation of CAPM- β												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.006*** (0.001)	0.017*** (0.002)	0.027*** (0.004)	0.036*** (0.009)	0.005*** (0.001)	0.017*** (0.002)	0.025*** (0.005)	0.035*** (0.009)	0.010*** (0.001)	0.021*** (0.004)	0.046*** (0.005)	0.062*** (0.008)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.014*** (0.001)	-0.043*** (0.007)	-0.067*** (0.010)	-0.091*** (0.018)	-0.016*** (0.002)	-0.050*** (0.003)	-0.086*** (0.006)	-0.151*** (0.030)	-0.017*** (0.002)	-0.037*** (0.004)	-0.089*** (0.006)	-0.156*** (0.026)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.001 (0.002)	-0.001 (0.007)	-0.010 (0.011)	-0.022 (0.019)	0.003 (0.002)	0.004 (0.003)	0.006 (0.006)	0.000 (0.031)	0.007*** (0.002)	0.005 (0.005)	0.033*** (0.007)	0.042* (0.024)
Intercept	0.005*** (0.001)	0.017*** (0.004)	0.038*** (0.007)	0.090*** (0.015)	0.005*** (0.001)	0.017*** (0.003)	0.039*** (0.007)	0.098*** (0.015)	0.002 (0.002)	0.014*** (0.005)	0.022*** (0.008)	0.078*** (0.014)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.014	0.049	0.067	0.067	0.015	0.055	0.073	0.113	0.013	0.034	0.057	0.094

Table IA4
Pooled Regression: Alternative Rolling Estimation of CAPM- β (Cont.)

Panel B. 60-month rolling estimation of CAPM- β

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.006*** (0.001)	0.016*** (0.003)	0.026*** (0.005)	0.033*** (0.010)	0.005*** (0.001)	0.017*** (0.002)	0.024*** (0.005)	0.033*** (0.010)	0.009*** (0.001)	0.020*** (0.004)	0.045*** (0.005)	0.058*** (0.008)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.015*** (0.001)	-0.042*** (0.006)	-0.066*** (0.009)	-0.091*** (0.017)	-0.017*** (0.002)	-0.050*** (0.003)	-0.086*** (0.006)	-0.152*** (0.031)	-0.018*** (0.002)	-0.039*** (0.004)	-0.089*** (0.006)	-0.148*** (0.023)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.002 (0.001)	-0.002 (0.006)	-0.010 (0.010)	-0.023 (0.018)	0.004** (0.002)	0.004 (0.003)	0.007 (0.006)	0.002 (0.032)	0.008*** (0.001)	0.007 (0.004)	0.033*** (0.007)	0.034 (0.021)
Intercept	0.005*** (0.001)	0.018*** (0.004)	0.040*** (0.008)	0.093*** (0.016)	0.005*** (0.001)	0.018*** (0.004)	0.040*** (0.008)	0.100*** (0.016)	0.002 (0.001)	0.015*** (0.005)	0.023*** (0.008)	0.081*** (0.015)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.014	0.049	0.067	0.067	0.016	0.055	0.073	0.113	0.013	0.035	0.056	0.092

Panel C. 120-month rolling estimation of CAPM- β

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.007*** (0.001)	0.019*** (0.004)	0.034*** (0.007)	0.050*** (0.017)	0.007*** (0.001)	0.021*** (0.003)	0.033*** (0.007)	0.049*** (0.015)	0.012*** (0.001)	0.027*** (0.004)	0.055*** (0.007)	0.080*** (0.014)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.014*** (0.003)	-0.036*** (0.012)	-0.051*** (0.014)	-0.051*** (0.018)	-0.015*** (0.002)	-0.046*** (0.009)	-0.060*** (0.012)	-0.069** (0.035)	-0.019*** (0.002)	-0.041*** (0.006)	-0.086*** (0.012)	-0.128*** (0.032)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.001 (0.003)	-0.009 (0.012)	-0.027** (0.013)	-0.065*** (0.019)	0.001 (0.002)	-0.001 (0.008)	-0.021** (0.011)	-0.085*** (0.030)	0.009*** (0.002)	0.008 (0.007)	0.029** (0.013)	0.013 (0.029)
Intercept	0.004** (0.002)	0.014*** (0.005)	0.031*** (0.010)	0.075*** (0.022)	0.004** (0.002)	0.013*** (0.005)	0.031*** (0.009)	0.082*** (0.020)	-0.000 (0.002)	0.008 (0.006)	0.012 (0.011)	0.058*** (0.021)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.014	0.047	0.065	0.066	0.015	0.054	0.069	0.106	0.014	0.035	0.056	0.090

Table IA5
Pooled Regression: Alternative High-Low Cutoff Value for the Expected Market Return

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return and $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns. $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below the trailing median of realized market returns and zero otherwise. The trailing median is the 48-, 60-, 120-month rolling median of realized market returns in Panels A to D, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. High-low cutoff value for the expected market return is the trailing median of realized market returns

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.003* (0.001)	0.020*** (0.006)	0.050*** (0.014)	0.030 (0.025)	0.017*** (0.003)	0.051*** (0.010)	0.103*** (0.014)	0.133*** (0.027)	0.010*** (0.003)	0.039*** (0.007)	0.080*** (0.016)	0.060*** (0.012)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.002 (0.002)	-0.021*** (0.007)	-0.055** (0.021)	-0.028 (0.043)	-0.018*** (0.004)	-0.056*** (0.015)	-0.116*** (0.021)	-0.159*** (0.045)	-0.011** (0.004)	-0.046*** (0.008)	-0.093*** (0.021)	-0.070*** (0.026)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.002 (0.002)	0.002 (0.009)	0.039 (0.026)	0.022 (0.048)	0.007 (0.005)	0.025* (0.015)	0.076*** (0.022)	0.108** (0.043)	0.003 (0.004)	0.006 (0.006)	0.035* (0.021)	0.009 (0.024)
Intercept	0.008*** (0.002)	0.017** (0.008)	0.006 (0.019)	0.068** (0.030)	0.000 (0.004)	-0.003 (0.011)	-0.028* (0.017)	-0.003 (0.029)	0.002 (0.003)	0.010* (0.005)	0.002 (0.016)	0.075*** (0.012)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.001	0.007	0.006	0.001	0.003	0.020	0.030	0.031	0.004	0.044	0.051	0.024

Table IA5
Pooled Regression: Alternative High-Low Cutoff Value for the Expected Market Return (Cont.)

Panel B. High-low cutoff value for the expected market return is the 48-month rolling median of realized market returns

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.004 (0.002)	0.009 (0.006)	0.007 (0.010)	0.025* (0.015)	0.004** (0.002)	0.009 (0.007)	0.015* (0.009)	0.033** (0.014)	0.003 (0.002)	0.012** (0.006)	0.016 (0.010)	0.046*** (0.014)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.007*** (0.002)	-0.014*** (0.003)	-0.005 (0.007)	-0.035*** (0.012)	-0.006*** (0.001)	-0.015*** (0.005)	-0.020*** (0.008)	-0.046*** (0.011)	-0.005*** (0.002)	-0.019*** (0.004)	-0.022*** (0.009)	-0.073*** (0.012)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.003 (0.003)	-0.005 (0.004)	-0.005 (0.009)	0.069*** (0.017)	-0.001 (0.002)	-0.010 (0.007)	0.004 (0.011)	0.059*** (0.018)	-0.004* (0.002)	-0.007 (0.006)	-0.005 (0.010)	0.052*** (0.015)
Intercept	0.009*** (0.003)	0.025*** (0.006)	0.044*** (0.011)	0.046*** (0.017)	0.009*** (0.002)	0.029*** (0.008)	0.040*** (0.011)	0.052*** (0.017)	0.009*** (0.002)	0.024*** (0.007)	0.044*** (0.012)	0.061*** (0.016)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.007	0.010	0.002	0.006	0.005	0.015	0.004	0.003	0.007	0.020	0.011	0.008

Panel C. High-low cutoff value for the expected market return is the 60-month rolling median of realized market returns

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.004*** (0.001)	0.013** (0.005)	0.012 (0.008)	0.030** (0.014)	0.004** (0.002)	0.010* (0.005)	0.017* (0.009)	0.042*** (0.014)	0.006*** (0.002)	0.014*** (0.005)	0.026*** (0.008)	0.041*** (0.011)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.008*** (0.002)	-0.025*** (0.004)	-0.018** (0.008)	-0.048*** (0.017)	-0.008*** (0.002)	-0.020*** (0.005)	-0.031*** (0.010)	-0.074*** (0.016)	-0.011*** (0.002)	-0.026*** (0.004)	-0.046*** (0.011)	-0.072*** (0.021)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.001 (0.002)	-0.005 (0.006)	-0.015 (0.011)	0.013 (0.023)	-0.000 (0.002)	-0.006 (0.006)	-0.008 (0.013)	0.043* (0.022)	-0.002 (0.002)	-0.006 (0.005)	-0.007 (0.013)	-0.015 (0.026)
Intercept	0.009*** (0.002)	0.027*** (0.007)	0.052*** (0.010)	0.086*** (0.016)	0.008*** (0.002)	0.028*** (0.006)	0.051*** (0.011)	0.072*** (0.017)	0.008*** (0.003)	0.025*** (0.006)	0.048*** (0.010)	0.105*** (0.013)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.007	0.026	0.013	0.009	0.006	0.018	0.020	0.011	0.015	0.030	0.039	0.047

Table IA5
Pooled Regression: Alternative High-Low Cutoff Value for the Expected Market Return (Cont.)

Panel D. High-low cutoff value for the expected market return is the 120-month rolling median of realized market returns

		Aggregation method for the expected market return											
		Median				Mean				PCA			
		h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
	$\beta_{i,t}$	0.003*** (0.001)	0.022*** (0.003)	0.024*** (0.006)	0.030*** (0.010)	0.005*** (0.001)	0.020*** (0.003)	0.026*** (0.005)	0.037*** (0.009)	0.010*** (0.001)	0.023*** (0.004)	0.048*** (0.007)	0.043*** (0.012)
	$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.005*** (0.002)	-0.035*** (0.008)	-0.038** (0.017)	-0.048* (0.028)	-0.009*** (0.003)	-0.034*** (0.008)	-0.047*** (0.017)	-0.068* (0.035)	-0.014*** (0.002)	-0.032*** (0.004)	-0.072*** (0.013)	-0.071* (0.042)
	$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.001 (0.002)	0.007 (0.008)	0.011 (0.018)	0.010 (0.028)	0.001 (0.003)	0.007 (0.008)	0.012 (0.017)	0.021 (0.035)	0.007*** (0.002)	0.008*** (0.003)	0.032*** (0.012)	0.026 (0.042)
	Intercept	0.007*** (0.001)	0.014*** (0.004)	0.033*** (0.008)	0.084*** (0.015)	0.006*** (0.001)	0.016*** (0.003)	0.037*** (0.006)	0.083*** (0.012)	0.001 (0.001)	0.011*** (0.004)	0.016** (0.007)	0.072*** (0.013)
	N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
	R^2	0.004	0.028	0.013	0.011	0.007	0.025	0.021	0.018	0.007	0.021	0.033	0.018

Table IA6
Pooled Regression: Inflation-Based High-Low Cutoff Value for the Expected Market Return

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < fitted]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < fitted]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return and $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns. $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < fitted]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below the fitted value of the current inflation rate obtained from a regression of historical expected market returns on historical inflation rates, and zero otherwise. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.005*** (0.001)	0.015*** (0.003)	0.028*** (0.005)	0.034*** (0.009)	0.004*** (0.001)	0.015*** (0.003)	0.028*** (0.004)	0.033*** (0.007)	0.006*** (0.002)	0.022*** (0.003)	0.050*** (0.006)	0.070*** (0.008)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < fitted]}$	-0.010*** (0.001)	-0.031*** (0.004)	-0.058*** (0.009)	-0.087*** (0.018)	-0.011*** (0.001)	-0.038*** (0.002)	-0.078*** (0.010)	-0.107*** (0.028)	-0.009*** (0.002)	-0.032*** (0.004)	-0.077*** (0.009)	-0.153*** (0.035)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < fitted]}$	-0.001 (0.001)	-0.001 (0.005)	0.003 (0.009)	-0.013 (0.019)	-0.001 (0.001)	0.002 (0.002)	0.008 (0.008)	-0.015 (0.026)	-0.000 (0.003)	0.002 (0.002)	0.025*** (0.007)	0.036 (0.033)
Intercept	0.005*** (0.001)	0.017*** (0.004)	0.032*** (0.007)	0.087*** (0.015)	0.006*** (0.002)	0.016*** (0.004)	0.035*** (0.007)	0.093*** (0.013)	0.005** (0.002)	0.014*** (0.003)	0.019*** (0.007)	0.069*** (0.012)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.010	0.029	0.039	0.053	0.012	0.035	0.060	0.076	0.008	0.029	0.050	0.100

Table IA7
Pooled Regressions Using Rolling Estimation of the Market Return Predictive Regressions

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return and $\beta_{i,t}$ is the CAPM- β . $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below its trailing median and zero otherwise. The individual market return forecasts described in Equations (3) and (4) are estimated over 48-, 60-, and 120-month rolling windows in Panels A to C, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. The individual forecast prior to aggregation are estimated out-of-sample using a 60-months rolling window. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. 48-month rolling market return predictive regression

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.004*** (0.001)	0.010*** (0.003)	0.019*** (0.007)	0.037*** (0.010)	0.001 (0.001)	0.010*** (0.002)	0.014** (0.006)	0.036*** (0.010)	0.001 (0.001)	0.005** (0.002)	0.005 (0.006)	0.028** (0.012)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.006*** (0.002)	-0.016*** (0.004)	-0.028*** (0.007)	-0.051*** (0.010)	-0.001 (0.002)	-0.018*** (0.003)	-0.020*** (0.005)	-0.055*** (0.009)	0.000 (0.001)	-0.006 (0.006)	-0.004 (0.006)	-0.048*** (0.013)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.003 (0.002)	-0.018*** (0.005)	-0.027*** (0.008)	-0.070*** (0.014)	-0.008*** (0.002)	-0.021*** (0.003)	-0.042*** (0.008)	-0.074*** (0.015)	-0.004** (0.001)	-0.015** (0.006)	-0.041*** (0.008)	-0.064*** (0.017)
Intercept	0.006*** (0.002)	0.022*** (0.005)	0.043*** (0.009)	0.100*** (0.017)	0.009*** (0.002)	0.024*** (0.003)	0.049*** (0.008)	0.102*** (0.017)	0.008*** (0.002)	0.024*** (0.004)	0.055*** (0.009)	0.106*** (0.019)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.006	0.027	0.034	0.078	0.006	0.034	0.040	0.087	0.001	0.011	0.021	0.068

Table IA7
Pooled Regressions Using Rolling Estimation of the Market Return Predictive Regressions (Cont.)

Panel B. 60-month rolling market return predictive regression												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.001 (0.002)	0.006 (0.005)	0.011 (0.009)	0.037*** (0.012)	0.003*** (0.001)	0.010*** (0.003)	0.009* (0.005)	0.025** (0.010)	-0.003 (0.002)	-0.002 (0.002)	0.009 (0.006)	0.024** (0.012)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.001 (0.002)	-0.007 (0.005)	-0.010 (0.008)	-0.046*** (0.011)	-0.005*** (0.001)	-0.015** (0.006)	-0.009 (0.006)	-0.035*** (0.010)	0.006*** (0.002)	0.008* (0.004)	-0.007 (0.007)	-0.031** (0.015)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.006** (0.002)	-0.022*** (0.007)	-0.035*** (0.011)	-0.049*** (0.016)	-0.006*** (0.001)	-0.019*** (0.006)	-0.041*** (0.007)	-0.086*** (0.014)	-0.010*** (0.003)	-0.027*** (0.005)	-0.040*** (0.008)	-0.066*** (0.017)
Intercept	0.008*** (0.002)	0.025*** (0.006)	0.048*** (0.011)	0.092*** (0.018)	0.008*** (0.001)	0.023*** (0.004)	0.052*** (0.008)	0.111*** (0.017)	0.011*** (0.003)	0.030*** (0.004)	0.052*** (0.009)	0.107*** (0.019)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.003	0.019	0.022	0.049	0.009	0.029	0.028	0.075	0.001	0.009	0.024	0.050

Panel C. 120-month rolling market return predictive regression												
Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_{i,t}$	0.000 (0.003)	0.001 (0.005)	0.010 (0.009)	0.011 (0.019)	-0.001 (0.002)	0.003 (0.003)	0.019*** (0.004)	0.023*** (0.007)	0.003 (0.002)	0.011*** (0.003)	0.028*** (0.003)	0.047*** (0.008)
$\beta_{i,t} \times \mathbb{I}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	0.002 (0.004)	0.006 (0.006)	-0.006 (0.007)	-0.000 (0.019)	0.006** (0.003)	0.000 (0.002)	-0.028*** (0.010)	-0.041** (0.018)	-0.004* (0.002)	-0.017*** (0.002)	-0.051*** (0.013)	-0.101*** (0.036)
$\mathbb{I}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.006 (0.004)	-0.024*** (0.007)	-0.025*** (0.008)	-0.049*** (0.018)	-0.011*** (0.003)	-0.028*** (0.004)	-0.029*** (0.009)	-0.073*** (0.017)	-0.000 (0.002)	-0.005** (0.002)	0.001 (0.012)	0.012 (0.037)
Intercept	0.008*** (0.003)	0.026*** (0.006)	0.045*** (0.010)	0.098*** (0.020)	0.010*** (0.002)	0.027*** (0.004)	0.042*** (0.006)	0.103*** (0.012)	0.006*** (0.002)	0.019*** (0.004)	0.035*** (0.005)	0.077*** (0.013)
N	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350	5,460	5,440	5,410	5,350
R^2	0.002	0.007	0.011	0.011	0.002	0.016	0.036	0.061	0.002	0.013	0.037	0.052

Table IA8
Pooled Regression: International Markets

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is portfolio i excess return and $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns. $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected excess market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 2, 6, 12$, is below its trailing median and zero otherwise. Panels A to C consider the Fama-French 25 size-and-book-to-market portfolios in Asian markets excluding Japan, developed markets excluding the U.S., and European markets, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected excess market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Robust standard errors clustered by portfolio and date are presented in parentheses and *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The out-of-sample period is from July 1990 to December 2022.

Panel A. Asian markets (excl. Japan)

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.044*** (0.006)	0.138*** (0.021)	0.209*** (0.037)	0.238*** (0.070)	0.077*** (0.011)	0.174*** (0.029)	0.234*** (0.044)	0.213*** (0.068)	0.069*** (0.009)	0.159*** (0.022)	0.312*** (0.048)	0.358*** (0.076)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.082*** (0.012)	-0.230*** (0.039)	-0.314*** (0.055)	-0.235*** (0.086)	-0.110*** (0.017)	-0.257*** (0.047)	-0.349*** (0.067)	-0.305*** (0.107)	-0.091*** (0.014)	-0.215*** (0.033)	-0.400*** (0.067)	-0.386*** (0.090)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.091*** (0.013)	0.234*** (0.041)	0.337*** (0.057)	0.263*** (0.087)	0.102*** (0.017)	0.248*** (0.047)	0.369*** (0.068)	0.395*** (0.111)	0.083*** (0.014)	0.188*** (0.034)	0.368*** (0.067)	0.377*** (0.092)
Intercept	-0.040*** (0.006)	-0.111*** (0.021)	-0.161*** (0.036)	-0.138* (0.071)	-0.064*** (0.011)	-0.141*** (0.029)	-0.183*** (0.042)	-0.137** (0.068)	-0.057*** (0.009)	-0.119*** (0.023)	-0.237*** (0.047)	-0.239*** (0.077)
N	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950
R^2	0.014	0.020	0.018	0.009	0.022	0.027	0.021	0.026	0.015	0.030	0.033	0.014

Table IA8
Pooled Regression: International Markets (Cont.)

Panel B. Developed markets (excl. U.S.)

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.002 (0.004)	0.018 (0.020)	-0.036 (0.029)	-0.189** (0.095)	0.016** (0.008)	0.042* (0.025)	0.003 (0.040)	-0.171* (0.087)	0.009 (0.006)	0.038* (0.020)	0.032 (0.054)	-0.161 (0.109)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.033*** (0.009)	-0.124*** (0.034)	-0.170*** (0.034)	-0.306*** (0.069)	-0.050*** (0.013)	-0.155*** (0.038)	-0.227*** (0.051)	-0.364*** (0.065)	-0.037*** (0.009)	-0.144*** (0.028)	-0.244*** (0.068)	-0.326*** (0.097)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.038*** (0.009)	0.115*** (0.033)	0.171*** (0.035)	0.304*** (0.071)	0.045*** (0.012)	0.140*** (0.037)	0.222*** (0.050)	0.385*** (0.066)	0.034*** (0.008)	0.122*** (0.028)	0.231*** (0.067)	0.314*** (0.098)
Intercept	0.002 (0.005)	0.008 (0.020)	0.083*** (0.029)	0.286*** (0.094)	-0.006 (0.008)	-0.012 (0.025)	0.047 (0.039)	0.258*** (0.086)	-0.001 (0.006)	-0.003 (0.020)	0.022 (0.054)	0.263** (0.109)
N	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950
R^2	0.006	0.012	0.015	0.045	0.008	0.019	0.019	0.051	0.004	0.024	0.021	0.046

Panel C. European markets

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1 (1)	h=3 (2)	h=6 (3)	h=12 (4)	h=1 (5)	h=3 (6)	h=6 (7)	h=12 (8)	h=1 (9)	h=3 (10)	h=6 (11)	h=12 (12)
$\beta_{i,t}$	0.003 (0.003)	0.011 (0.013)	-0.015 (0.019)	-0.190** (0.079)	0.009*** (0.003)	0.030** (0.014)	0.002 (0.023)	-0.158** (0.068)	0.001 (0.004)	0.023 (0.016)	0.001 (0.035)	-0.204** (0.082)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.015** (0.007)	-0.061*** (0.020)	-0.123*** (0.029)	-0.164*** (0.045)	-0.025*** (0.006)	-0.090*** (0.020)	-0.144*** (0.032)	-0.238*** (0.058)	-0.009* (0.005)	-0.081*** (0.020)	-0.132*** (0.038)	-0.124** (0.056)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.022*** (0.007)	0.051** (0.020)	0.121*** (0.031)	0.161*** (0.048)	0.019*** (0.006)	0.075*** (0.021)	0.142*** (0.033)	0.261*** (0.062)	0.007 (0.005)	0.058*** (0.020)	0.124*** (0.039)	0.127** (0.059)
Intercept	0.001 (0.004)	0.019 (0.013)	0.071*** (0.020)	0.306*** (0.083)	0.001 (0.003)	0.004 (0.013)	0.055** (0.023)	0.262*** (0.070)	0.008* (0.004)	0.015 (0.016)	0.057 (0.037)	0.317*** (0.086)
N	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950	6,125	6,125	6,100	5,950
R^2	0.004	0.004	0.006	0.026	0.004	0.010	0.008	0.030	0.001	0.015	0.007	0.025

Table IA9
Out-of-Sample R^2 for Beta-Sorted Portfolios with Alternative Rolling Estimation of CAPM- β

This table reports the out-of-sample R^2 (OOS- R^2) for the ten beta-sorted portfolios. The OOS- R^2 for portfolio i is computed as follows

$$\text{OOS-}R_i^2 = 1 - \frac{\sum_{t=1}^N (r_{i,t+h} - \mathbb{E}_t(r_{i,t+h}))^2}{\sum_{t=1}^N (r_{i,t+h} - \bar{r}_{i,t+h})^2},$$

where $r_{i,t+h}$ is portfolio i excess return over horizon h , $\bar{r}_{i,t+h}$ is the average historical return of portfolio i over horizon h prior to month t , and $\mathbb{E}_t(r_{i,t+h})$ for $h = 1, 3, 6, 12$, is the expected return of portfolio i computed as in Equation (9). In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Panels A to C report the results using a 48-,60-,120-month rolling CAPM- β , respectively. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. 48-month rolling CAPM- β												
Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.007**	0.024***	0.027***	0.019***	0.008**	0.035***	0.049***	0.049***	0.011***	0.055***	0.072***	0.060***
2	0.005*	0.016***	0.009***	-0.019***	0.006**	0.027***	0.031***	0.009***	0.012***	0.055***	0.063***	0.032***
3	0.007**	0.025***	0.025***	0.009***	0.008**	0.036***	0.047***	0.036***	0.012***	0.062***	0.083***	0.068***
4	0.009***	0.032***	0.036***	0.026***	0.010***	0.042***	0.059***	0.054***	0.015***	0.070***	0.099***	0.094***
5	0.009***	0.032***	0.034***	0.010***	0.010***	0.043***	0.057***	0.037***	0.013***	0.071***	0.100***	0.080***
6	0.008***	0.030***	0.025***	-0.013***	0.010***	0.043***	0.053***	0.019***	0.013***	0.072***	0.096***	0.066***
7	0.009***	0.036***	0.037***	0.006***	0.011***	0.052***	0.069***	0.045***	0.013***	0.078***	0.110***	0.090***
8	0.010***	0.037***	0.030***	-0.029***	0.012***	0.054***	0.066***	0.013***	0.013***	0.080***	0.110***	0.064***
9	0.011***	0.042***	0.041***	-0.008***	0.014***	0.061***	0.079***	0.037***	0.012***	0.080***	0.123***	0.092***
High β	0.009***	0.041***	0.031***	-0.054***	0.012***	0.063***	0.077***	0.001***	0.005**	0.067***	0.112***	0.047***

Table IA9
Out-of-Sample R^2 for Beta-Sorted Portfolios with Alternative Rolling Estimation of CAPM- β (Cont.)

Panel B. 60-month rolling CAPM- β

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.007**	0.025***	0.028***	0.021***	0.008**	0.037***	0.052***	0.052***	0.010**	0.057***	0.075***	0.066***
2	0.005*	0.016***	0.009***	-0.019***	0.006**	0.028***	0.031***	0.010***	0.012***	0.056***	0.064***	0.036***
3	0.007**	0.025***	0.024***	0.008***	0.008**	0.035***	0.046***	0.036***	0.013***	0.062***	0.082***	0.070***
4	0.009***	0.031***	0.036***	0.027***	0.010***	0.041***	0.058***	0.055***	0.014***	0.069***	0.097***	0.094***
5	0.009***	0.031***	0.032***	0.008***	0.010***	0.041***	0.055***	0.035***	0.013***	0.070***	0.098***	0.078***
6	0.008**	0.029***	0.024***	-0.011***	0.009***	0.042***	0.052***	0.022***	0.013***	0.071***	0.095***	0.066***
7	0.009***	0.036***	0.038***	0.007***	0.012***	0.052***	0.070***	0.045***	0.013***	0.077***	0.111***	0.089***
8	0.010***	0.037***	0.031***	-0.026***	0.012***	0.054***	0.066***	0.015***	0.013***	0.079***	0.109***	0.064***
9	0.011***	0.043***	0.042***	-0.009***	0.014***	0.061***	0.080***	0.035***	0.012***	0.080***	0.121***	0.092***
High β	0.009***	0.041***	0.031***	-0.059***	0.013***	0.064***	0.077***	-0.003***	0.005***	0.070***	0.112***	0.045***

Panel C. 120-month rolling CAPM- β

Aggregation method for the expected market return												
	Median				Mean				PCA			
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.006*	0.022***	0.026***	0.022***	0.007**	0.033***	0.048***	0.051***	0.011***	0.054***	0.076***	0.071***
2	0.004*	0.016***	0.012***	-0.006***	0.006*	0.026***	0.033***	0.020***	0.012***	0.052***	0.066***	0.045***
3	0.007**	0.024***	0.026***	0.020***	0.008**	0.034***	0.048***	0.046***	0.013***	0.060***	0.083***	0.076***
4	0.009***	0.030***	0.038***	0.038***	0.010***	0.041***	0.060***	0.065***	0.014***	0.067***	0.098***	0.098***
5	0.008**	0.029***	0.031***	0.010***	0.009**	0.040***	0.055***	0.038***	0.013***	0.069***	0.096***	0.077***
6	0.008**	0.029***	0.026***	-0.004***	0.009**	0.042***	0.054***	0.029***	0.012***	0.070***	0.094***	0.067***
7	0.009***	0.035***	0.035***	0.005***	0.011***	0.051***	0.068***	0.044***	0.012***	0.076***	0.105***	0.084***
8	0.010***	0.038***	0.033***	-0.021***	0.012***	0.055***	0.068***	0.020***	0.011***	0.078***	0.105***	0.062***
9	0.011***	0.044***	0.043***	-0.002***	0.014***	0.063***	0.083***	0.045***	0.010***	0.079***	0.118***	0.094***
High β	0.010***	0.045***	0.036***	-0.044***	0.014***	0.069***	0.084***	0.014***	0.004**	0.071***	0.111***	0.058***

Table IA10
Out-of-Sample R^2 for Beta-Sorted Portfolios with Rolling Estimation of the Intertemporal Hedging
Component, $c_{i,t+h}$

This table reports the out-of-sample R^2 (OOS- R^2) for the ten beta-sorted portfolios. The OOS- R^2 for portfolio i is computed as follows

$$\text{OOS-}R_i^2 = 1 - \frac{\sum_{t=1}^N (r_{i,t+h} - \mathbb{E}_t(r_{i,t+h}))^2}{\sum_{t=1}^N (r_{i,t+h} - \bar{r}_{i,t+h})^2},$$

where $r_{i,t+h}$ is portfolio i excess return over horizon h , $\bar{r}_{i,t+h}$ is the average historical return of portfolio i over horizon h prior to month t , and $\mathbb{E}_t(r_{i,t+h})$ for $h = 1, 3, 6, 12$, is the expected return of portfolio i computed as in Equation (9). Panels A to C present the results with a 48-, 60-, 120-month rolling estimation of the intertemporal hedging component $c_{i,t+h}$, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. 48-month rolling estimation of $c_{i,t+h}$

Forecast aggregation method												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	-0.016	-0.014***	-0.026***	-0.035***	-0.016	-0.006***	-0.008***	-0.009***	-0.017	0.004***	0.006***	-0.007***
2	-0.008	0.005***	0.001***	0.014***	-0.006	0.014***	0.022***	0.040***	-0.001*	0.041***	0.055***	0.064***
3	-0.007	0.012***	0.017***	0.030***	-0.005	0.023***	0.040***	0.061***	-0.002*	0.047***	0.068***	0.073***
4	-0.006	0.016***	0.019***	0.022***	-0.005	0.026***	0.041***	0.047***	-0.002*	0.051***	0.076***	0.084***
5	-0.010	0.013***	0.020***	0.022***	-0.009	0.025***	0.046***	0.052***	-0.006*	0.051***	0.087***	0.098***
6	-0.012	0.011***	0.010***	-0.008***	-0.010	0.027***	0.043***	0.032***	-0.007	0.052***	0.079***	0.071***
7	-0.016	0.008***	0.002***	-0.032***	-0.012	0.029***	0.043***	0.020***	-0.010*	0.055***	0.080***	0.063***
8	-0.020	-0.002***	-0.015***	-0.059***	-0.015	0.021***	0.029***	-0.008***	-0.015*	0.041***	0.064***	0.059***
9	-0.024	-0.009***	-0.031***	-0.110***	-0.019	0.015***	0.016***	-0.059***	-0.022*	0.029***	0.050***	0.020***
High β	-0.044	-0.047***	-0.089***	-0.211***	-0.036	-0.010***	-0.018***	-0.128***	-0.044	-0.013***	0.007***	-0.049***

Table IA10
Out-of-Sample R^2 for Beta-Sorted Portfolios with Rolling Estimation of the Intertemporal Hedging
Component, $c_{i,t+h}$ (Cont.)

Panel B. 60-month rolling estimation of $c_{i,t+h}$

		Forecast aggregation method											
		Median				Mean				PCA			
		h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β		-0.018	-0.025*	-0.050***	-0.100***	-0.017	-0.017**	-0.032***	-0.070***	-0.021	-0.012***	-0.030***	-0.085***
	2	-0.008	-0.003***	-0.015***	-0.031***	-0.008	0.007***	0.006***	-0.002***	-0.004*	0.028***	0.029***	0.009***
	3	-0.010	-0.003***	-0.016***	-0.047***	-0.009	0.008***	0.007***	-0.014***	-0.007	0.027***	0.027***	-0.013***
	4	-0.008	0.004***	-0.008***	-0.039***	-0.007	0.014***	0.016***	-0.009***	-0.006	0.035***	0.044***	0.019***
	5	-0.010	0.006***	-0.003***	-0.038***	-0.008	0.018***	0.023***	-0.005***	-0.007	0.040***	0.056***	0.030***
	6	-0.013	-0.001***	-0.021***	-0.077***	-0.011	0.015***	0.013***	-0.033***	-0.010	0.036***	0.042***	-0.006***
	7	-0.016	-0.003***	-0.028***	-0.099***	-0.012	0.017***	0.013***	-0.044***	-0.013	0.037***	0.042***	-0.012***
	8	-0.016	-0.003***	-0.028***	-0.095***	-0.012	0.019***	0.016***	-0.042***	-0.014*	0.034***	0.044***	0.014***
	9	-0.019	-0.006***	-0.032***	-0.118***	-0.015	0.018***	0.015***	-0.063***	-0.021*	0.025***	0.041***	-0.003***
High β		-0.032	-0.028***	-0.069***	-0.200***	-0.025	0.006***	-0.001***	-0.118***	-0.037	-0.002***	0.015***	-0.076***

Panel C. 120-month rolling estimation of $c_{i,t+h}$

		Forecast aggregation method											
		Median				Mean				PCA			
		h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β		0.001	0.015***	0.015***	0.004***	0.001	0.025***	0.035***	0.038***	-0.001*	0.036***	0.042***	0.012***
	2	0.003*	0.017***	0.015***	0.011***	0.004*	0.028***	0.038***	0.043***	0.009***	0.056***	0.069***	0.050***
	3	0.001	0.014***	0.007***	-0.014***	0.002	0.026***	0.032***	0.020***	0.006**	0.054***	0.065***	0.028***
	4	0.003*	0.021***	0.019***	0.005***	0.004*	0.033***	0.044***	0.037***	0.010**	0.064***	0.088***	0.066***
	5	0.001	0.018***	0.010***	-0.026***	0.002*	0.031***	0.037***	0.008***	0.007*	0.063***	0.083***	0.040***
	6	-0.001	0.014***	-0.002***	-0.054***	0.002*	0.031***	0.032***	-0.009***	0.006**	0.064***	0.079***	0.021***
	7	-0.001	0.017***	0.002***	-0.055***	0.002*	0.039***	0.044***	0.001***	0.007**	0.073***	0.094***	0.033***
	8	-0.001*	0.018***	-0.001***	-0.067***	0.003*	0.041***	0.042***	-0.012***	0.007**	0.072***	0.094***	0.037***
	9	-0.003*	0.018***	0.000***	-0.083***	0.003*	0.046***	0.051***	-0.018***	0.004**	0.072***	0.102***	0.024***
High β		-0.016	-0.014***	-0.057***	-0.202**	-0.009	0.025***	0.012***	-0.111***	-0.012*	0.037***	0.051***	-0.089***

Table IA11
Out-of-Sample R^2 for Beta-Sorted Portfolios with Rolling Estimation of the Market Return Predictive Regressions

This table reports the out-of-sample R^2 (OOS- R^2) for the ten beta-sorted portfolios. The OOS- R^2 for portfolio i is computed as follows

$$\text{OOS-}R_i^2 = 1 - \frac{\sum_{t=1}^N (r_{i,t+h} - \mathbb{E}_t(r_{i,t+h}))^2}{\sum_{t=1}^N (r_{i,t+h} - \bar{r}_{i,t+h})^2},$$

where $r_{i,t+h}$ is portfolio i excess return over horizon h , $\bar{r}_{i,t+h}$ is the average historical return of portfolio i over horizon h prior to month t , and $\mathbb{E}_t(r_{i,t+h})$ for $h = 1, 3, 6, 12$, is the expected return of portfolio i computed as in Equation (9). The individual market return forecasts described in Equations (3) and (4) are estimated over 48-, 60-, and 120-month rolling windows in Panels A to C, respectively. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. 48-month rolling market return predictive regression												
Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.004*	0.039***	0.063***	0.093***	0.012	0.045***	0.067***	0.111***	-0.176	0.035***	0.031***	0.084***
2	0.002*	0.031***	0.049***	0.062***	0.013	0.039***	0.057***	0.083***	-0.079	0.032***	0.026***	0.064***
3	0.002*	0.037***	0.064***	0.094***	0.008	0.047***	0.070***	0.115***	-0.227	0.036***	0.034***	0.093***
4	0.003*	0.041***	0.073***	0.111***	0.005	0.052***	0.079***	0.135***	-0.300	0.036***	0.037***	0.115***
5	0.001*	0.040***	0.064***	0.087***	-0.002	0.050***	0.071***	0.110***	-0.386	0.022***	0.019***	0.088***
6	0.001*	0.040***	0.062***	0.079***	-0.008	0.054***	0.073***	0.110***	-0.480	0.022***	0.014***	0.078***
7	0.001**	0.047***	0.079***	0.104***	-0.009	0.061***	0.089***	0.139***	-0.513	0.015***	0.008***	0.092***
8	-0.000**	0.044***	0.067***	0.072***	-0.010	0.059***	0.080***	0.111***	-0.515	0.002***	-0.006***	0.057***
9	-0.008*	0.028***	0.047***	0.054***	-0.027	0.046***	0.061***	0.099***	-0.674	-0.022***	-0.031***	0.056***
High β	-0.014*	0.023***	0.034***	0.025***	-0.071	0.043***	0.053***	0.083***	-1.198	-0.075***	-0.086***	-0.001***

Table IA11
Out-of-Sample R^2 for Beta-Sorted Portfolios with Rolling Estimation of the Market Return Predictive Regressions (Cont.)

Panel B. 60-month rolling market return predictive regression												
Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.004*	0.038***	0.060***	0.079***	0.018*	0.049***	0.067***	0.104***	-0.062	0.043***	0.048***	0.072***
2	0.002*	0.029***	0.040***	0.043***	0.015*	0.040***	0.053***	0.076***	-0.016	0.038***	0.036***	0.049***
3	0.001	0.035***	0.056***	0.077***	0.016*	0.049***	0.068***	0.110***	-0.089	0.040***	0.045***	0.077***
4	0.002*	0.038***	0.065***	0.093***	0.016*	0.054***	0.078***	0.131***	-0.126	0.035***	0.046***	0.098***
5	-0.001*	0.036***	0.056***	0.068***	0.011*	0.052***	0.069***	0.106***	-0.173	0.017***	0.025***	0.065***
6	-0.002*	0.035***	0.051***	0.053***	0.008*	0.054***	0.070***	0.104***	-0.221	0.013***	0.021***	0.055***
7	-0.003	0.039***	0.063***	0.068***	0.007*	0.059***	0.082***	0.129***	-0.239	0.002***	0.014***	0.067***
8	-0.005	0.035***	0.052***	0.036***	0.005*	0.057***	0.075***	0.104***	-0.242	-0.014***	-0.004***	0.034***
9	-0.013	0.023***	0.041***	0.035***	-0.006*	0.047***	0.063***	0.107***	-0.330	-0.048***	-0.030***	0.036***
High β	-0.023	0.012***	0.025***	0.002***	-0.030	0.042***	0.055***	0.097***	-0.615	-0.123***	-0.084***	-0.016***

Panel C. 120-month rolling market return predictive regression												
Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.001	0.017***	0.032***	0.050***	-0.004	0.026***	0.047***	0.071***	-0.009	0.025***	0.026***	0.066***
2	-0.001	0.009***	0.012***	0.008***	-0.004	0.020***	0.031***	0.036***	-0.003	0.028***	0.029***	0.038***
3	-0.002	0.013***	0.027***	0.043***	-0.006	0.024***	0.045***	0.068***	-0.010	0.029***	0.040***	0.066***
4	-0.001	0.017***	0.034***	0.053***	-0.005	0.029***	0.056***	0.083***	-0.009	0.034***	0.048***	0.076***
5	-0.004	0.013***	0.027***	0.032***	-0.009	0.026***	0.050***	0.066***	-0.015	0.024***	0.032***	0.051***
6	-0.005	0.011***	0.018***	0.007***	-0.010	0.029***	0.050***	0.057***	-0.015	0.027***	0.031***	0.040***
7	-0.007	0.012***	0.024***	0.013***	-0.011	0.034***	0.063***	0.078***	-0.018	0.029***	0.038***	0.044***
8	-0.009	0.006***	0.010***	-0.031***	-0.012	0.033***	0.057***	0.049***	-0.019	0.024***	0.031***	0.022***
9	-0.014	0.002***	0.009***	-0.027***	-0.018	0.032***	0.062***	0.060***	-0.027	0.018***	0.029***	-0.003***
High β	-0.024	-0.013***	-0.011***	-0.088***	-0.028	0.029***	0.061***	0.043***	-0.051	-0.002***	0.006***	-0.041***

Table IA12
Out-of-Sample R^2 for Beta-Sorted Portfolios with the Conditional CAPM Specification ($c_{i,t+h} = 0, \forall i, t, h$)

This table reports the out-of-sample R^2 (OOS- R^2) for the ten beta-sorted portfolios. The OOS- R^2 for portfolio i is computed as follows

$$\text{OOS-}R_i^2 = 1 - \frac{\sum_{t=1}^N (r_{i,t+h} - \mathbb{E}_t(r_{i,t+h}))^2}{\sum_{t=1}^N (r_{i,t+h} - \bar{r}_{i,t+h})^2},$$

where $r_{i,t+h}$ is portfolio i excess return over horizon h , $\bar{r}_{i,t+h}$ is the average historical return of portfolio i over horizon h prior to month t , and $\mathbb{E}_t(r_{i,t+h})$ for $h = 1, 3, 6, 12$, is the expected return of portfolio i computed as in Equation (9) with $c_{i,t+h} = 0, \forall i, t, h$. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the median, mean, and PCA aggregation methods, respectively. Statistical significance based on [Clark and West \(2007\)](#) one-sided test at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. The out-of-sample period is from July 1977 to December 2022.

Aggregation method for the expected market return												
Median				Mean				PCA				
	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Low β	0.009***	0.025***	0.026***	0.004***	0.008**	0.031***	0.040***	0.022***	0.009***	0.048***	0.056***	0.021***
2	0.008**	0.022***	0.018***	-0.013***	0.008**	0.030***	0.035***	0.008***	0.013***	0.057***	0.066***	0.025***
3	0.009***	0.027***	0.027***	0.006***	0.009**	0.034***	0.044***	0.025***	0.013***	0.062***	0.080***	0.051***
4	0.010***	0.031***	0.035***	0.017***	0.011***	0.040***	0.055***	0.040***	0.015***	0.070***	0.097***	0.078***
5	0.013***	0.039***	0.045***	0.034***	0.013***	0.048***	0.067***	0.058***	0.017***	0.077***	0.110***	0.099***
6	0.015***	0.043***	0.050***	0.034***	0.016***	0.056***	0.078***	0.066***	0.019***	0.084***	0.119***	0.110***
7	0.015***	0.048***	0.059***	0.047***	0.018***	0.064***	0.092***	0.088***	0.020***	0.089***	0.134***	0.131***
8	0.014***	0.044***	0.047***	0.019***	0.017***	0.063***	0.084***	0.066***	0.019***	0.086***	0.128***	0.122***
9	0.012***	0.040***	0.035***	-0.012***	0.016***	0.062***	0.078***	0.044***	0.018***	0.082***	0.130***	0.109***
High β	0.005***	0.021***	-0.007***	-0.096***	0.010***	0.049***	0.048***	-0.022***	0.008***	0.049***	0.090***	0.045***

Table IA13
Pooled Regression: Aggregated Price and Sentiment Forecasts of the Market Return

This table reports estimates from the following pooled regressions

$$r_{i,t+h} = a + b_1\beta_{it} + b_2\beta_{it} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + b_3\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]} + \epsilon_{i,t+h},$$

where $r_{i,t}$ is beta portfolio i excess return, $\mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$ corresponds to a dummy variable equal to one if the expected market return, $\mathbb{E}_t(r_{M,t+h})$ for $h = 1, 3, 6, 12$, is below its trailing median and zero otherwise, and $\beta_{i,t}$ is the CAPM- β estimated using the previous 24 monthly returns. In columns (1)-(4), (5)-(8), and (9)-(12), $\mathbb{E}_t(r_{M,t+h}) \geq 0$ is the non-negative expected market return defined in Section 3.2 and constructed using the mean aggregation method applied to the variance risk premium and short-interest in Panel A (i.e., price forecast) and to investor sentiment, managerial sentiment, and investor attention in Panel B (i.e., sentiment forecast). ***, **, and * corresponds to the 1, 5, and 10% statistical level, respectively. The out-of-sample period is from July 1977 to December 2022.

Panel A. Aggregated price forecast of the market return				
	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)
$\beta_{i,t}$	0.005** (0.002)	0.015*** (0.003)	0.027*** (0.006)	0.036*** (0.014)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.008*** (0.002)	-0.028*** (0.003)	-0.045*** (0.006)	-0.054*** (0.010)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	0.004 (0.003)	0.009* (0.004)	0.011 (0.008)	-0.005 (0.016)
Intercept	0.004* (0.002)	0.013*** (0.005)	0.029*** (0.009)	0.079*** (0.019)
N	5,460	5,440	5,410	5,350
R^2	0.003	0.013	0.019	0.022

Table IA13
Pooled Regression: Aggregated Price and Sentiment Forecasts of the Market Return

Panel B. Aggregated sentiment forecast of the market return				
	h=1	h=3	h=6	h=12
	(1)	(2)	(3)	(4)
$\beta_{i,t}$	0.001 (0.001)	0.003 (0.002)	0.013*** (0.004)	0.025*** (0.006)
$\beta_{i,t} \times \mathbb{1}_{[\mathbb{E}_t(r_{M,t+h}) < med]}$	-0.002 (0.002)	-0.003 (0.005)	-0.033** (0.015)	-0.080** (0.036)
$\mathbb{1}_{[\mathbb{E}_t(r_{M,t+1}) < med]}$	-0.003** (0.002)	-0.013*** (0.004)	0.001 (0.015)	0.000 (0.036)
Intercept	0.007*** (0.001)	0.023*** (0.003)	0.040*** (0.006)	0.089*** (0.012)
N	5,460	5,440	5,410	5,350
R^2	0.002	0.005	0.012	0.033